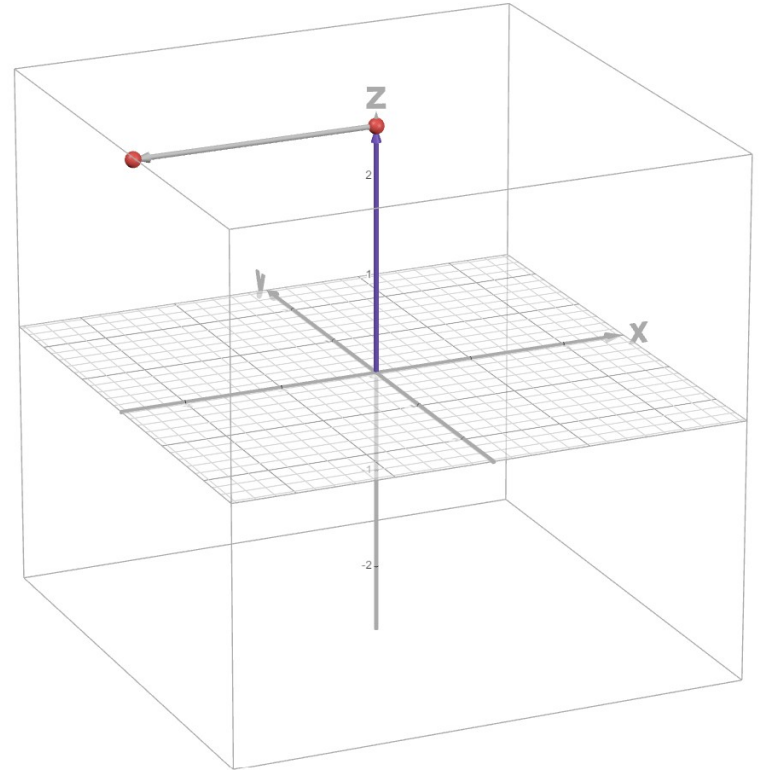
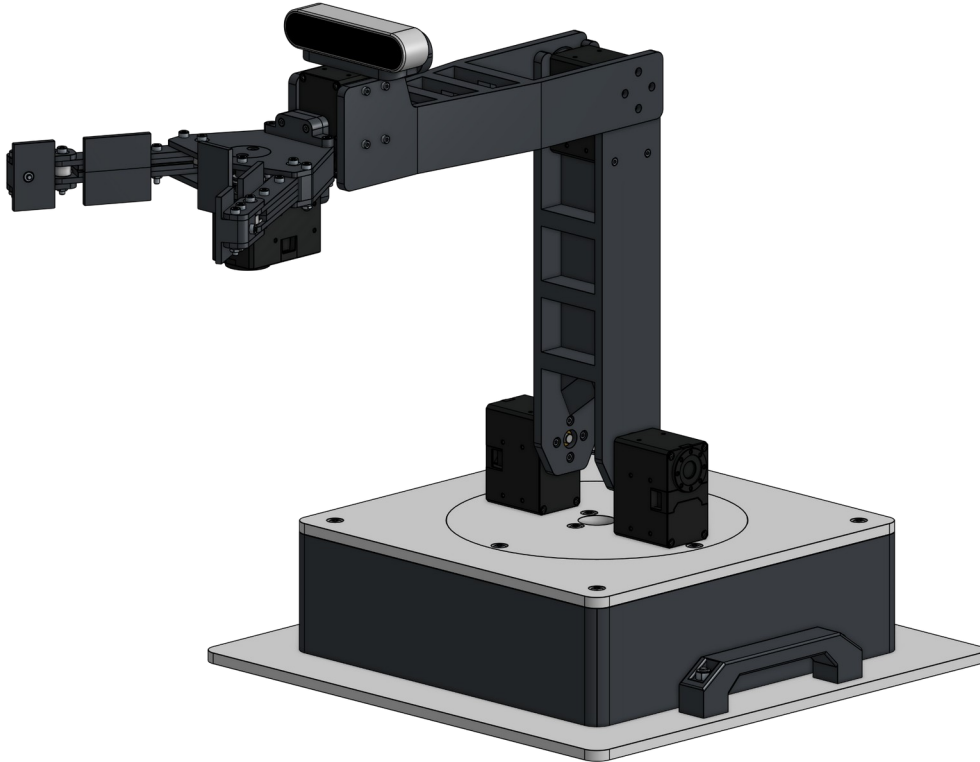


The Trigonometric Approach to Forward Kinematics

By Joseph Ruhan



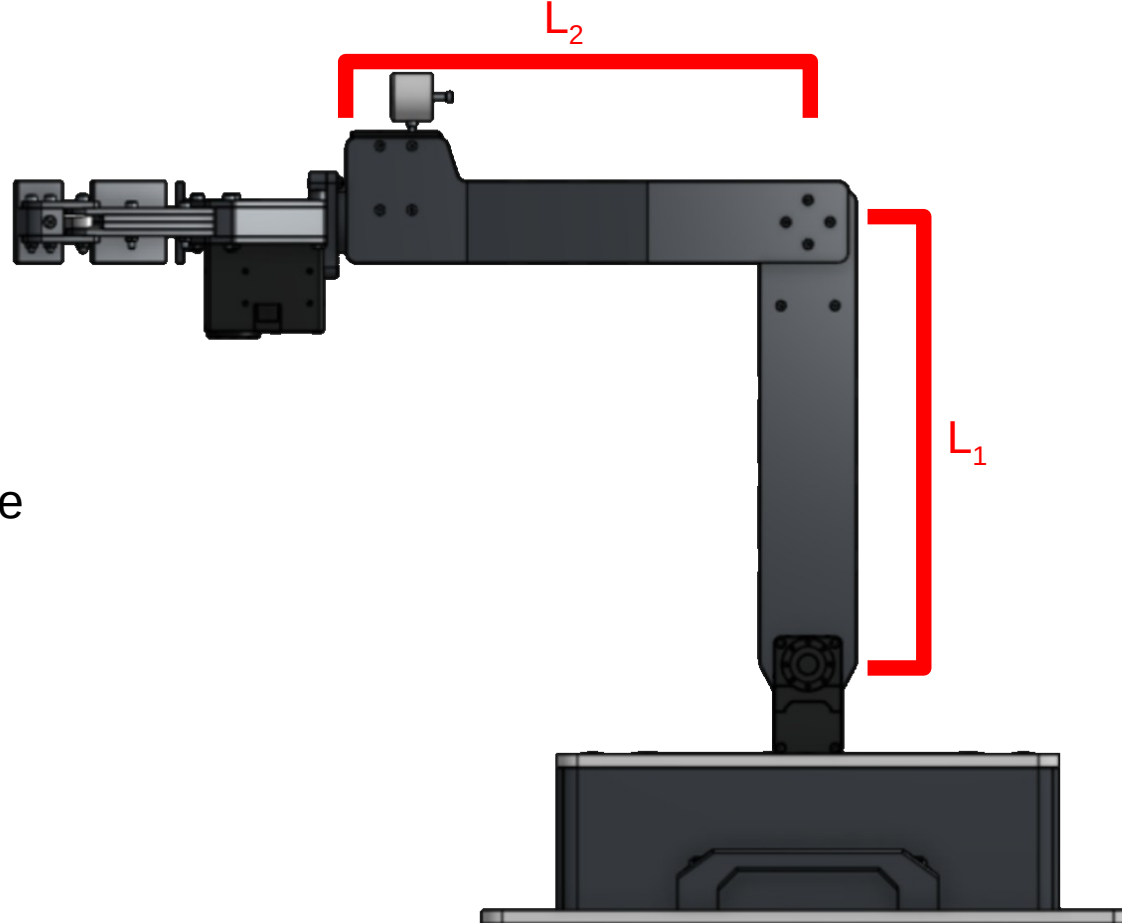
Forward

Where

$$L_1 = 250\text{mm}$$

$$L_2 = 250\text{mm}$$

These are always
known because they are
a constant



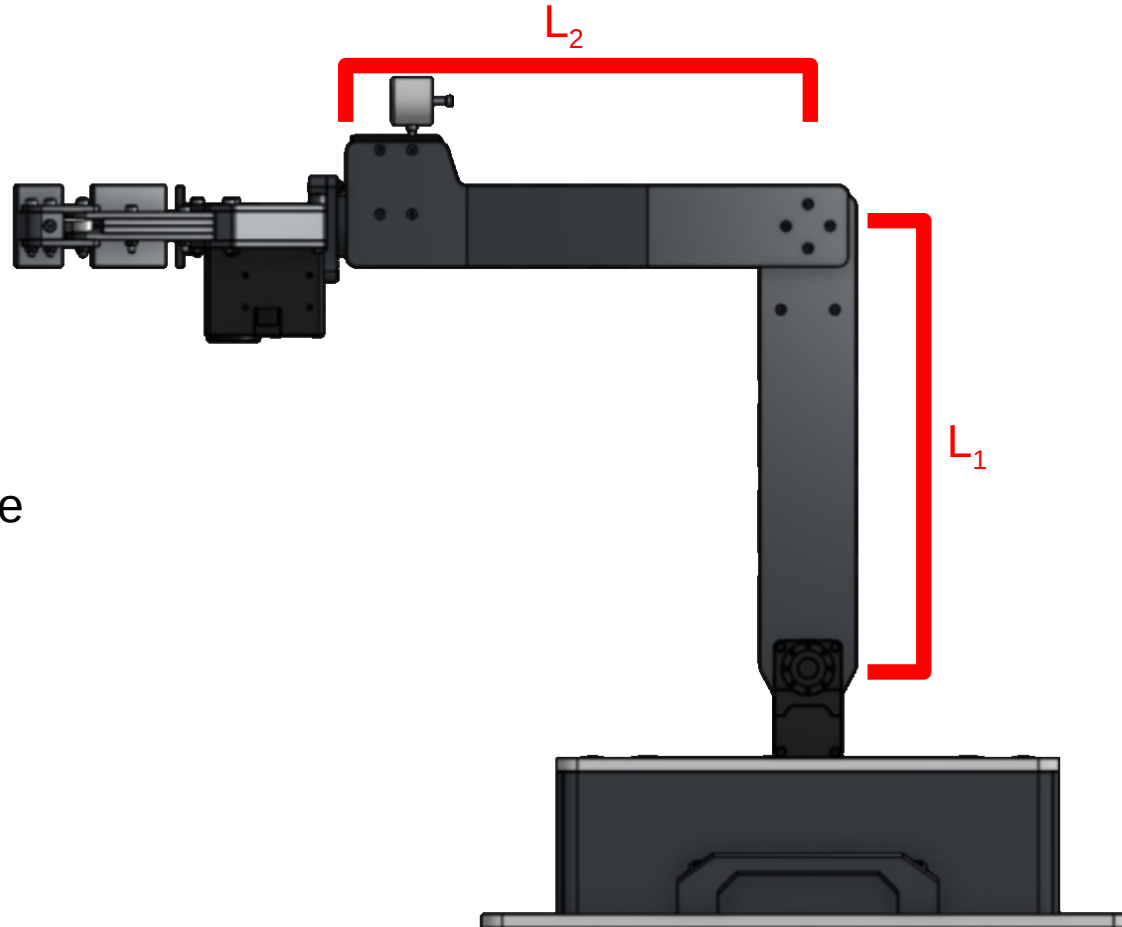
The arm can be described mathematically by defining its joints and links

Where

$$L_1 = 250\text{mm}$$

$$L_2 = 250\text{mm}$$

These are always known because they are a constant

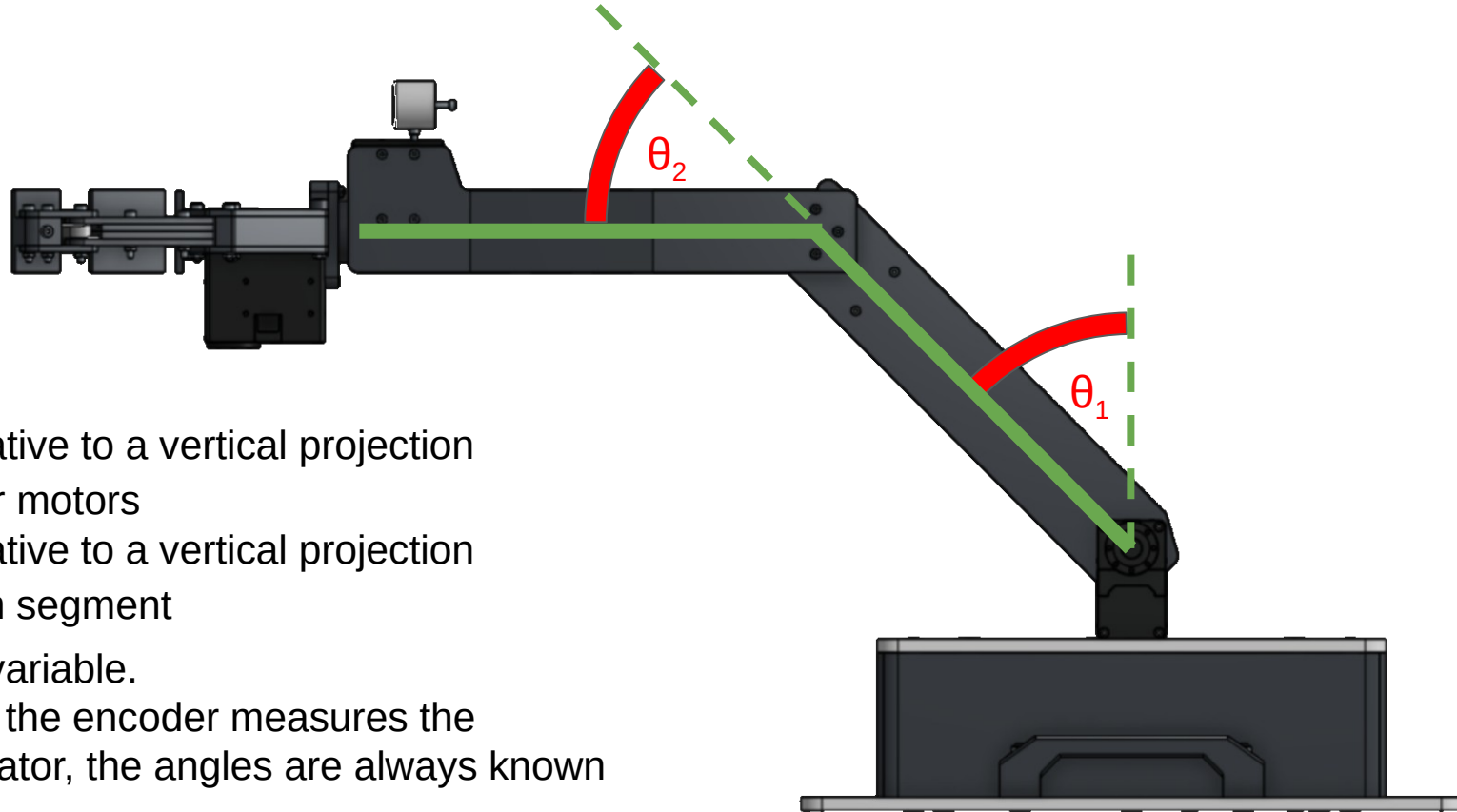


The arm can be described mathematically by defining its joints and links

In this example

$$\theta_1 = 45^\circ$$

$$\theta_2 = 45^\circ$$



θ_1 is measured relative to a vertical projection above the shoulder motors

θ_1 is measured relative to a vertical projection from the upper arm segment

These angles are variable.

However, because the encoder measures the rotation of the actuator, the angles are always known

The arm can therefore be represented as such:

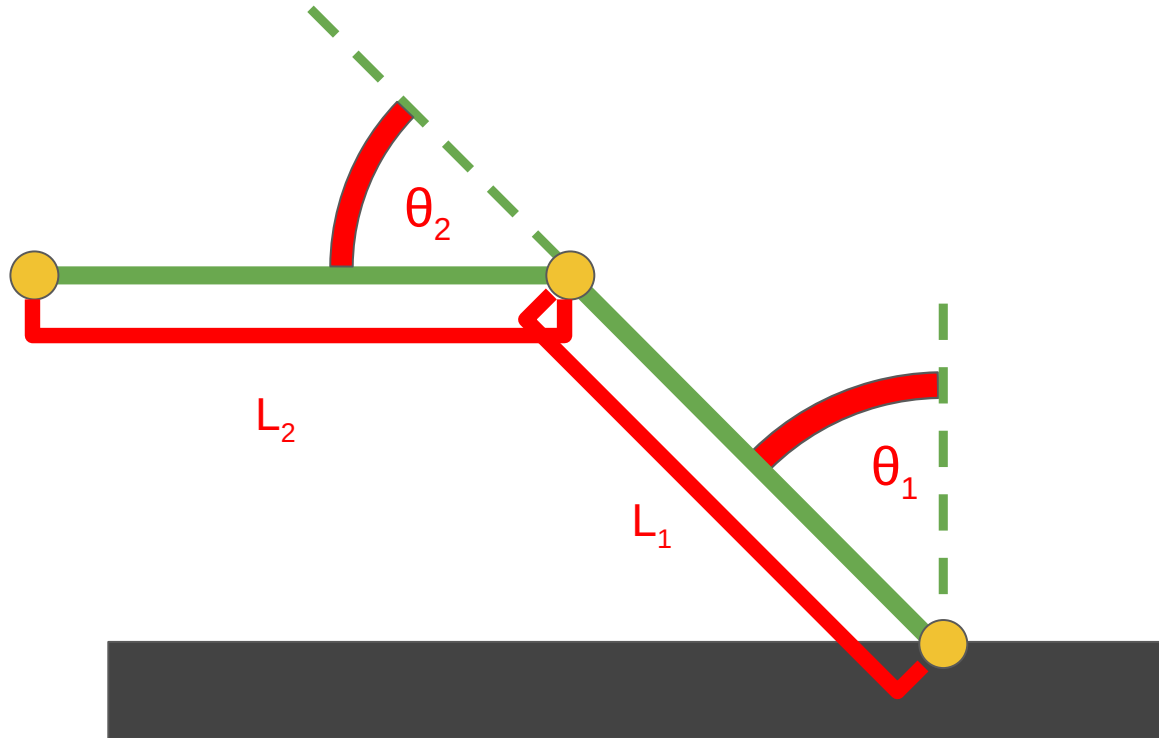
● = a joint

$L_1=250\text{mm}$

$L_2=250\text{mm}$

$\theta_1=45^\circ$

$\theta_2=45^\circ$



The arm can therefore be represented as such:

 = a joint

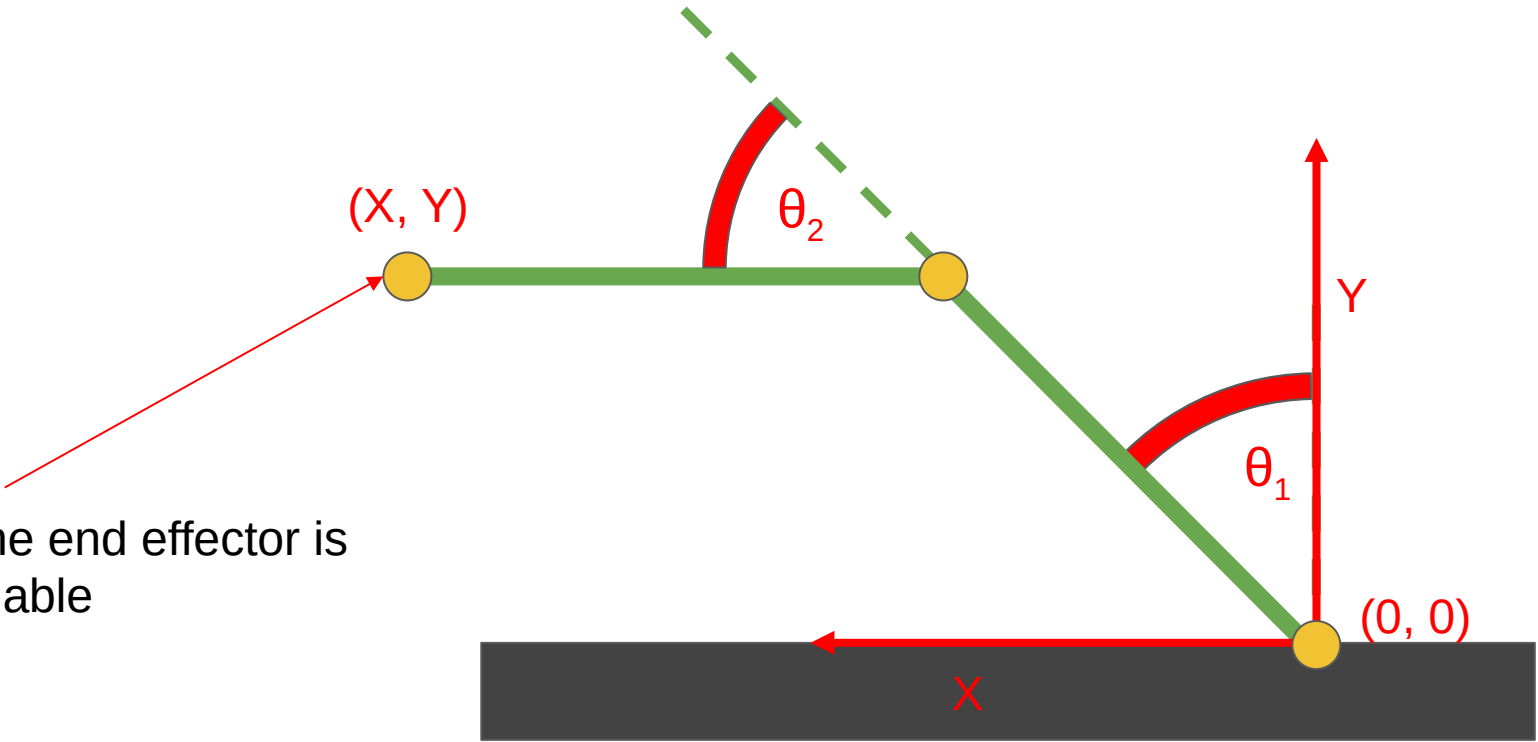
$L_1=250\text{mm}$

$L_2=250\text{mm}$

$\theta_1=45^\circ$

$\theta_2=45^\circ$

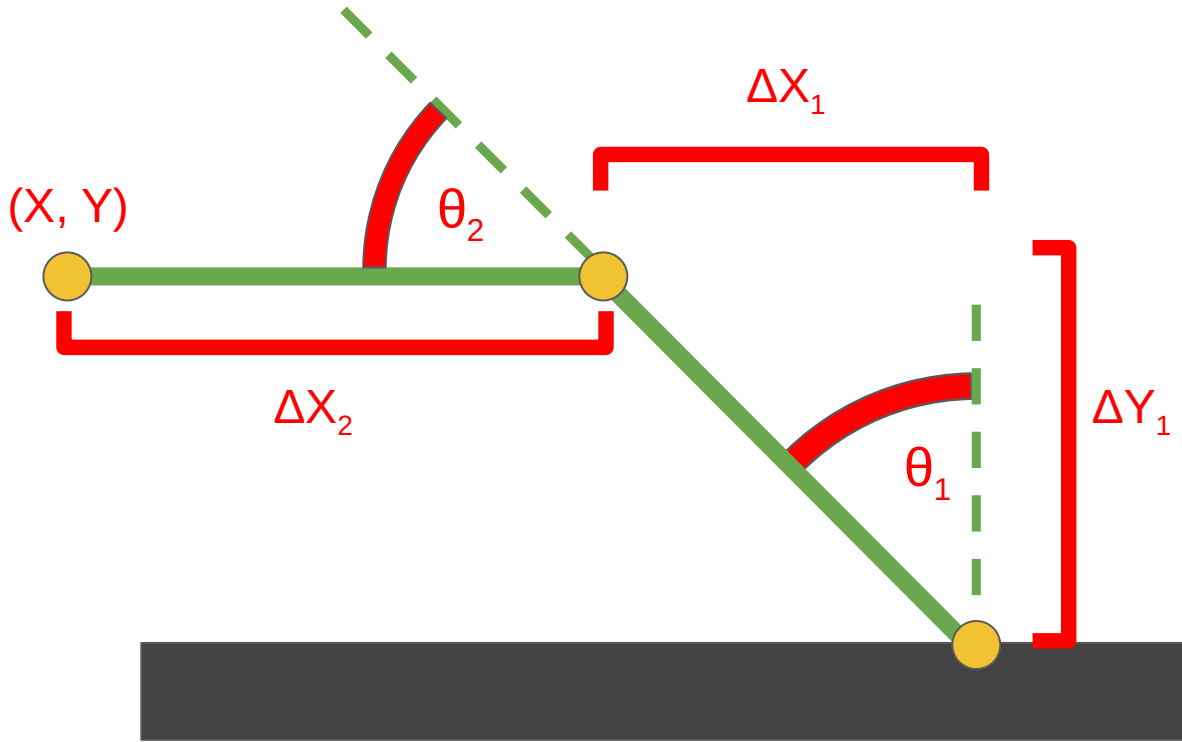
The location of the end effector is
the unknown variable



The arm can therefore be represented as such:

 = a joint

$L_1=250\text{mm}$
 $L_2=250\text{mm}$
 $\theta_1=45^\circ$
 $\theta_2=45^\circ$



The position of the end effector will be a sum of the vertical and horizontal displacement of each joint

The arm can therefore be represented as such:

 = a joint

$L_1=250\text{mm}$

$L_2=250\text{mm}$

$\theta_1=45^\circ$

$\theta_2=45^\circ$

(X, Y)

ΔX_2

θ_2

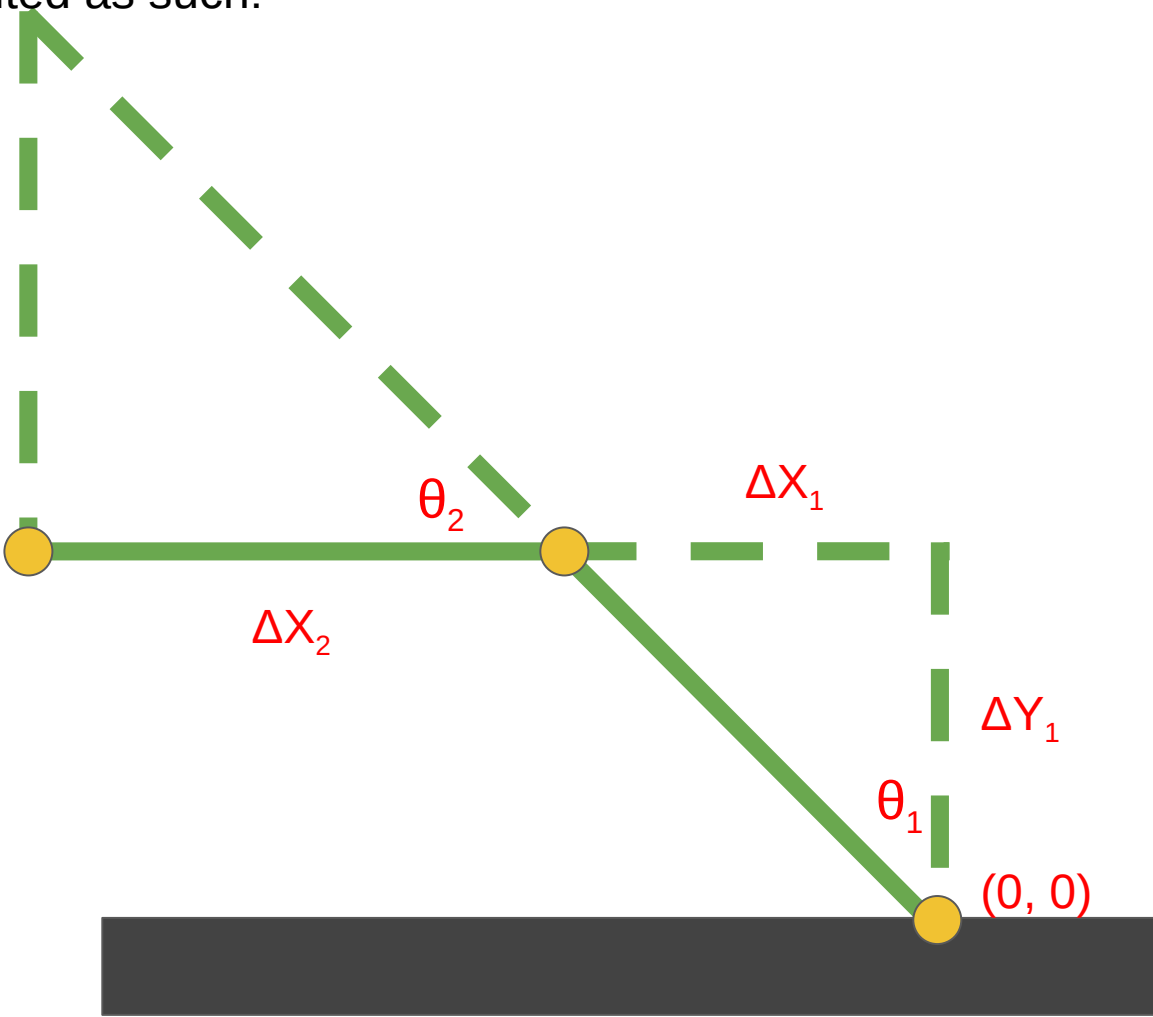
ΔX_1

ΔY_1

θ_1

$(0, 0)$

The displacement can be represented by triangles



To calculate the position of the end point:

$$L_1 = 250\text{mm}$$

$$L_2 = 250\text{mm}$$

$$\theta_1 = 45^\circ$$

$$\theta_2 = 45^\circ$$

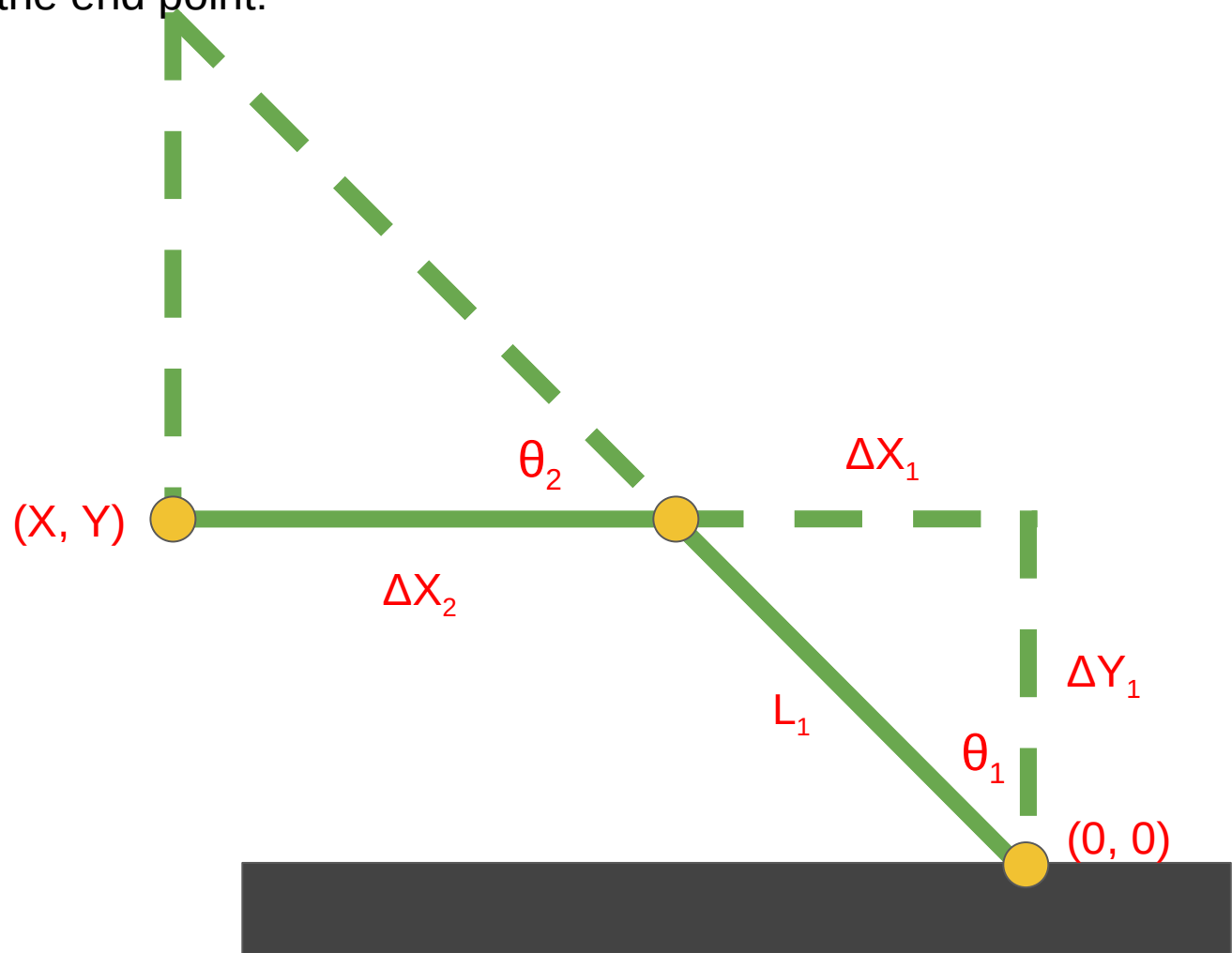
In this example, L_1 is the hypotenuse of a triangle.

Using $\sin(\theta_1)$:

$$\sin(\theta_1) = \Delta X_1 / L_1$$

$$L_1 \sin(\theta_1) = \Delta X_1$$

$$250 * (\sqrt{2}/2) = \Delta X_1$$



To calculate the position of the end point:

$$\Delta X_1 = 176.777\text{mm}$$

For $\cos(\theta_1)$:

$$\cos(\theta_1) = \Delta Y_1 / L_1$$

$$L_1 \sin(\theta_1) = \Delta Y_1$$

$$250 * (\sqrt{2}/2) = \Delta Y_1$$

$$\Delta Y_1 = 176.777\text{mm}$$

And $\cos(\theta_2)$:

$$\cos(\theta_2) = \Delta X_2 / L_2$$

$$L_2 \cos(\theta_1) = \Delta X_2$$

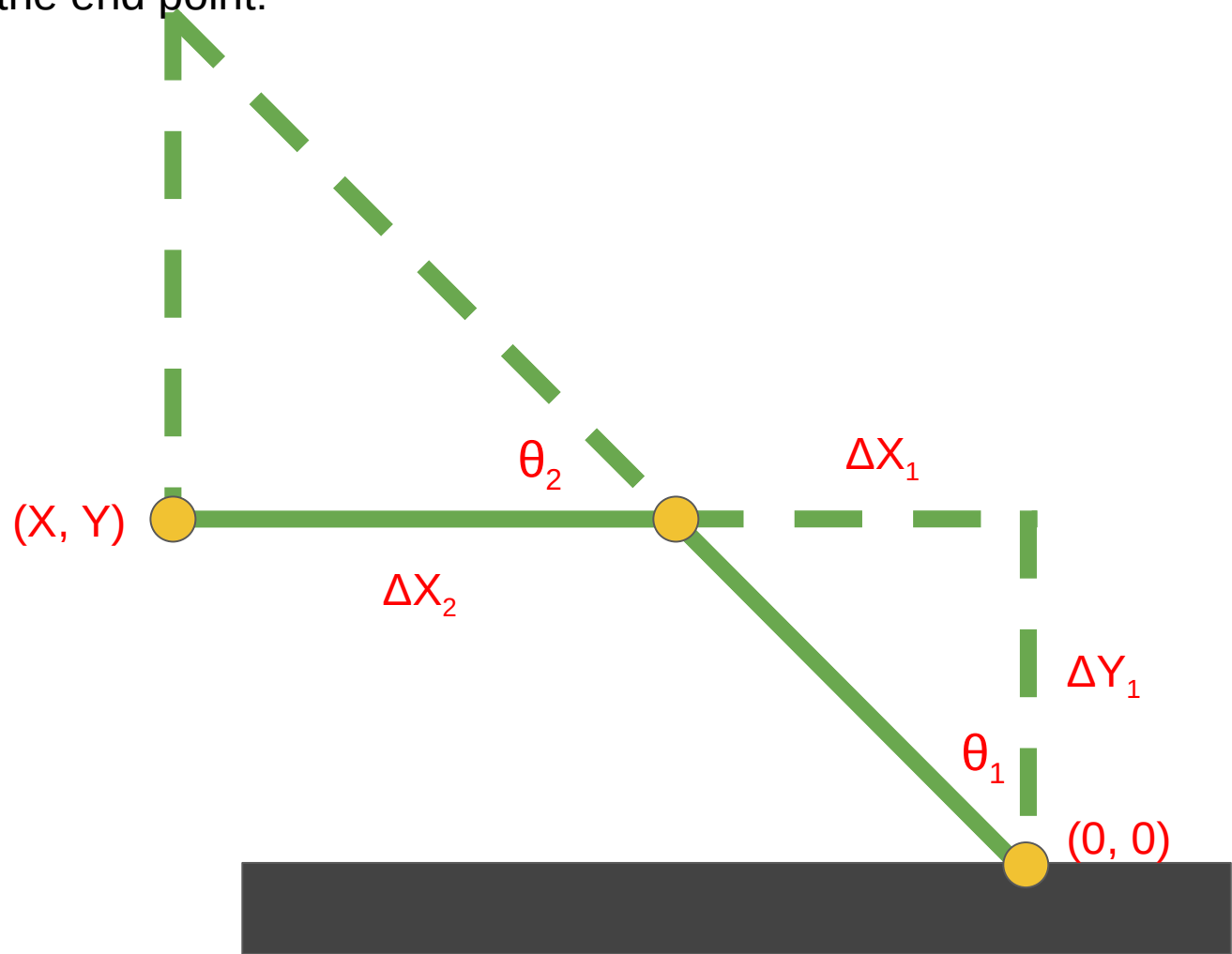
$$250 * (1) = \Delta X_2$$

$$\Delta X_2 = 250\text{mm}$$

And $\sin(\theta_2)$:

$$\cos(\theta_2) = \Delta Y_2 / L_2$$

$$L_2 \cos(\theta_1) = \Delta Y_2$$



To calculate the position of the end point:

$$\Delta X_1 = 176.777\text{mm}$$

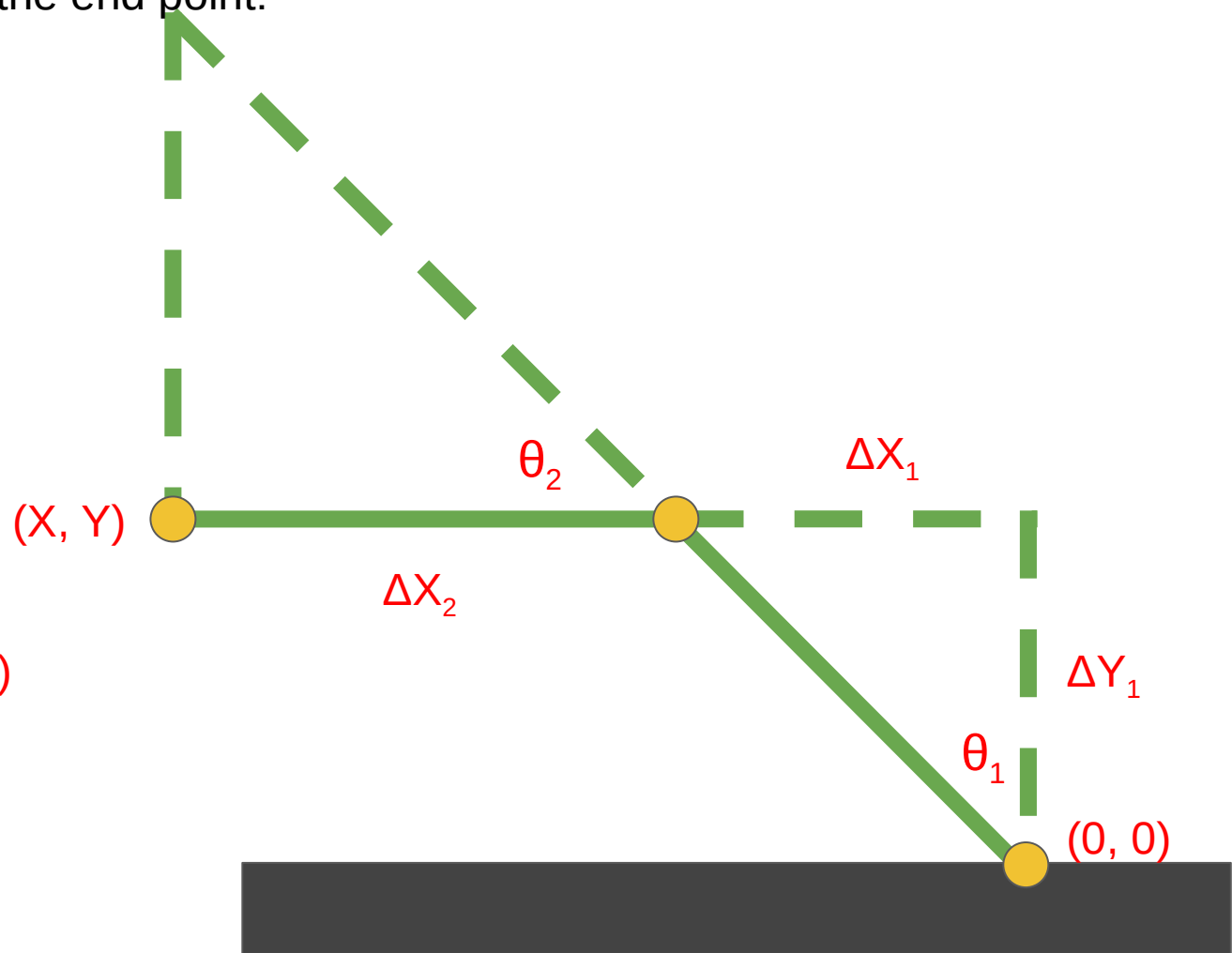
$$\Delta Y_1 = 176.777\text{mm}$$

$$\Delta X_2 = 250\text{mm}$$

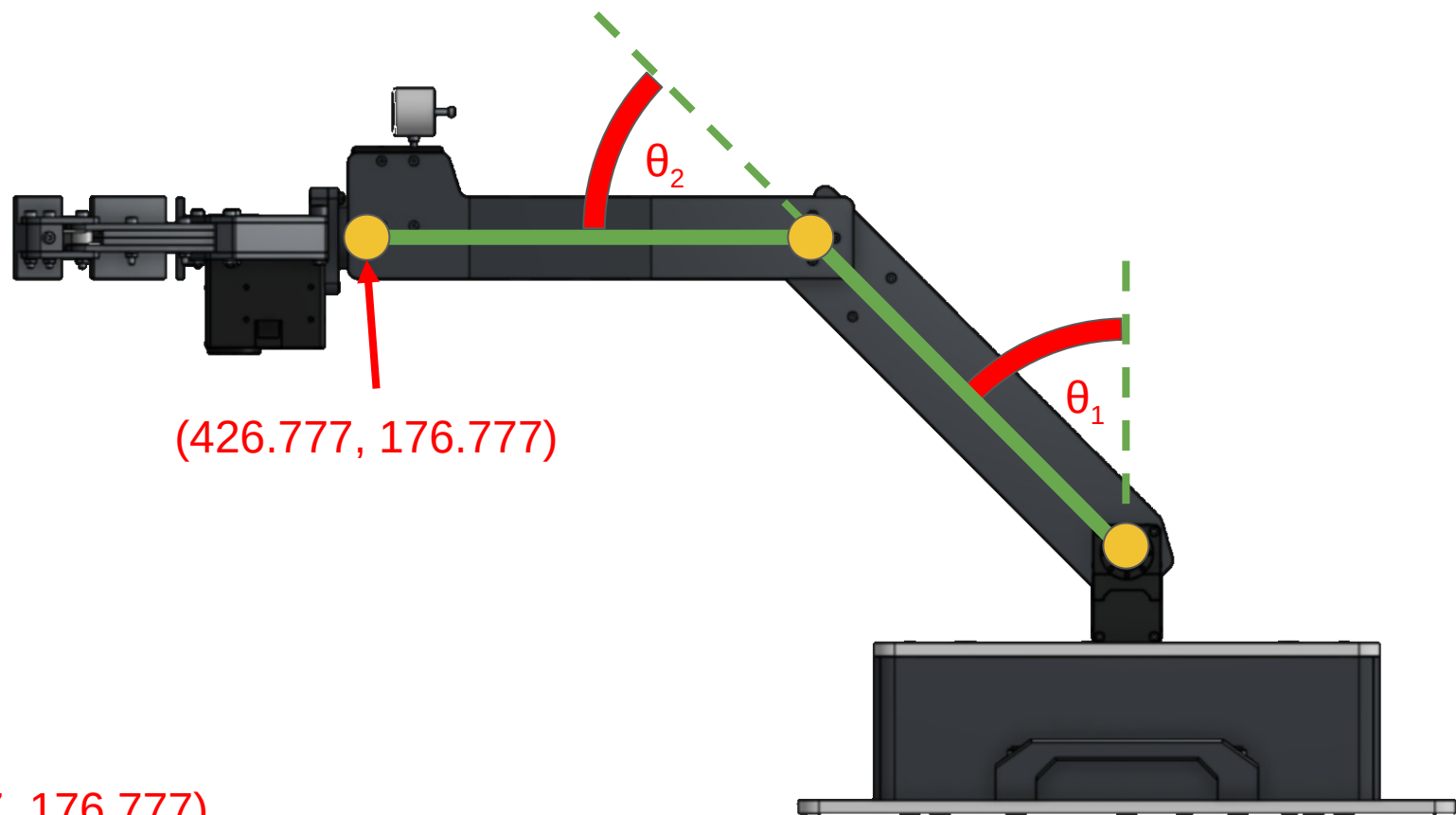
$$\Delta Y_2 = 0\text{mm}$$

$$(X, Y) = (\Delta X_1 + \Delta X_2, \Delta Y_1 + \Delta Y_2)$$

$$(X, Y) = (426.777, 176.777)$$



Completed Variables



- $L_1=250\text{mm}$
- $L_2=250\text{mm}$
- $\theta_1=45^\circ$
- $\theta_2=45^\circ$
- $(X,Y) = (426.777, 176.777)$

Try It Yourself:

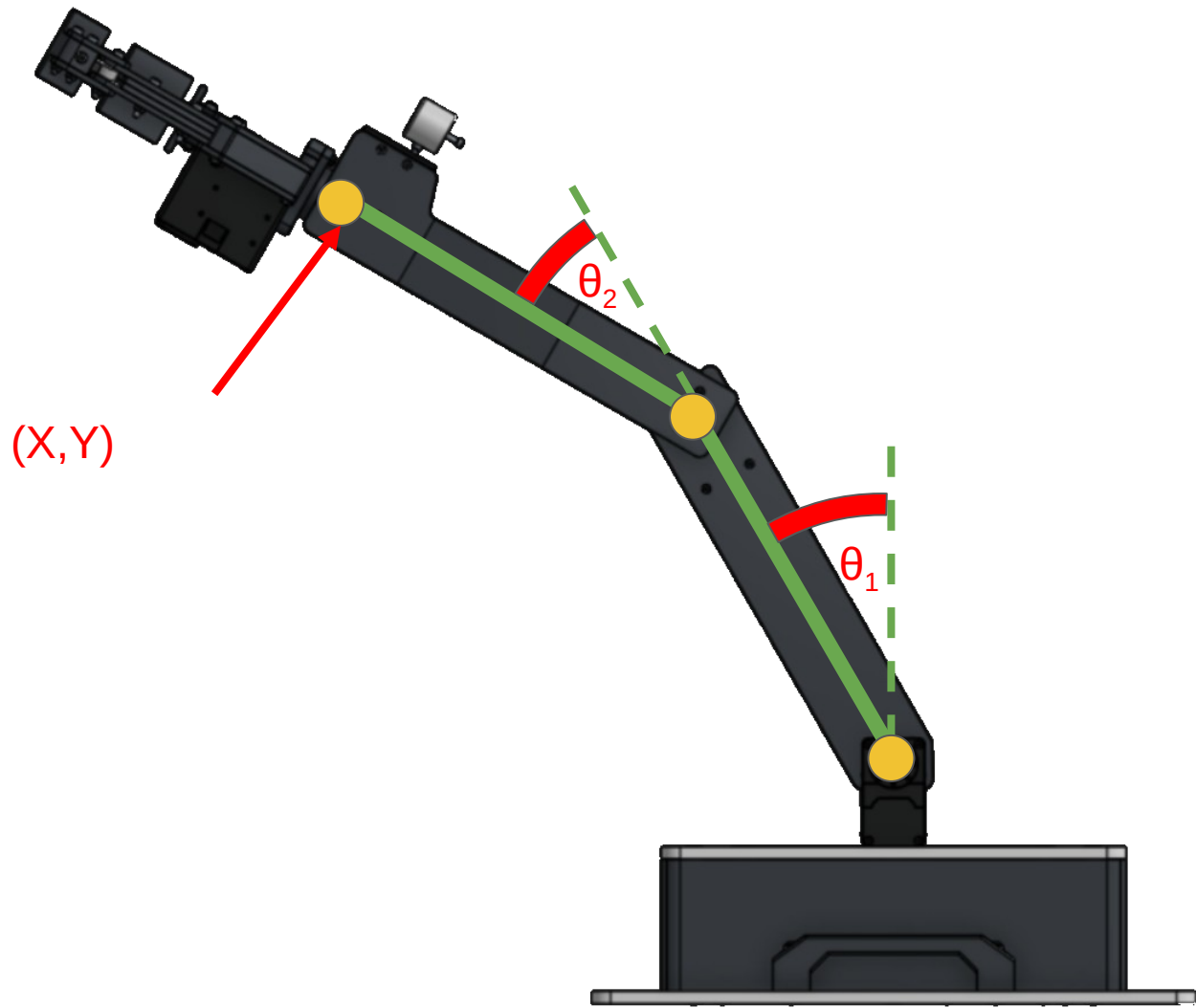
$L_1=250\text{mm}$

$L_2=250\text{mm}$

$\theta_1=30^\circ$

$\theta_2=30^\circ$

$(X,Y) = (?,?)$

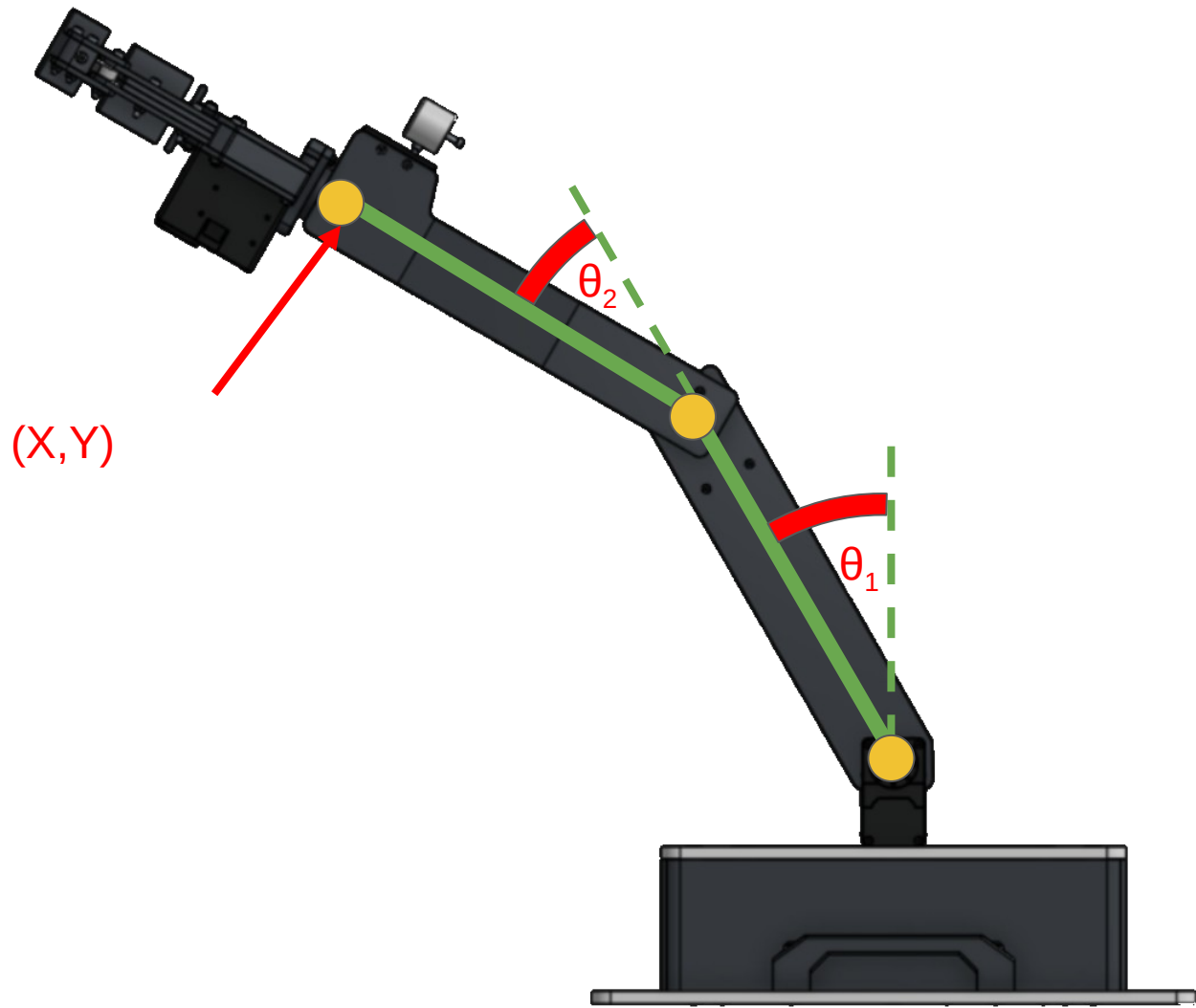


Try It Yourself:

$L_1=250\text{mm}$
 $L_2=250\text{mm}$
 $\theta_1=30^\circ$
 $\theta_2=30^\circ$
 $(X,Y) = (?,?)$

Solution:

Click To Reveal



Try It Yourself 2:

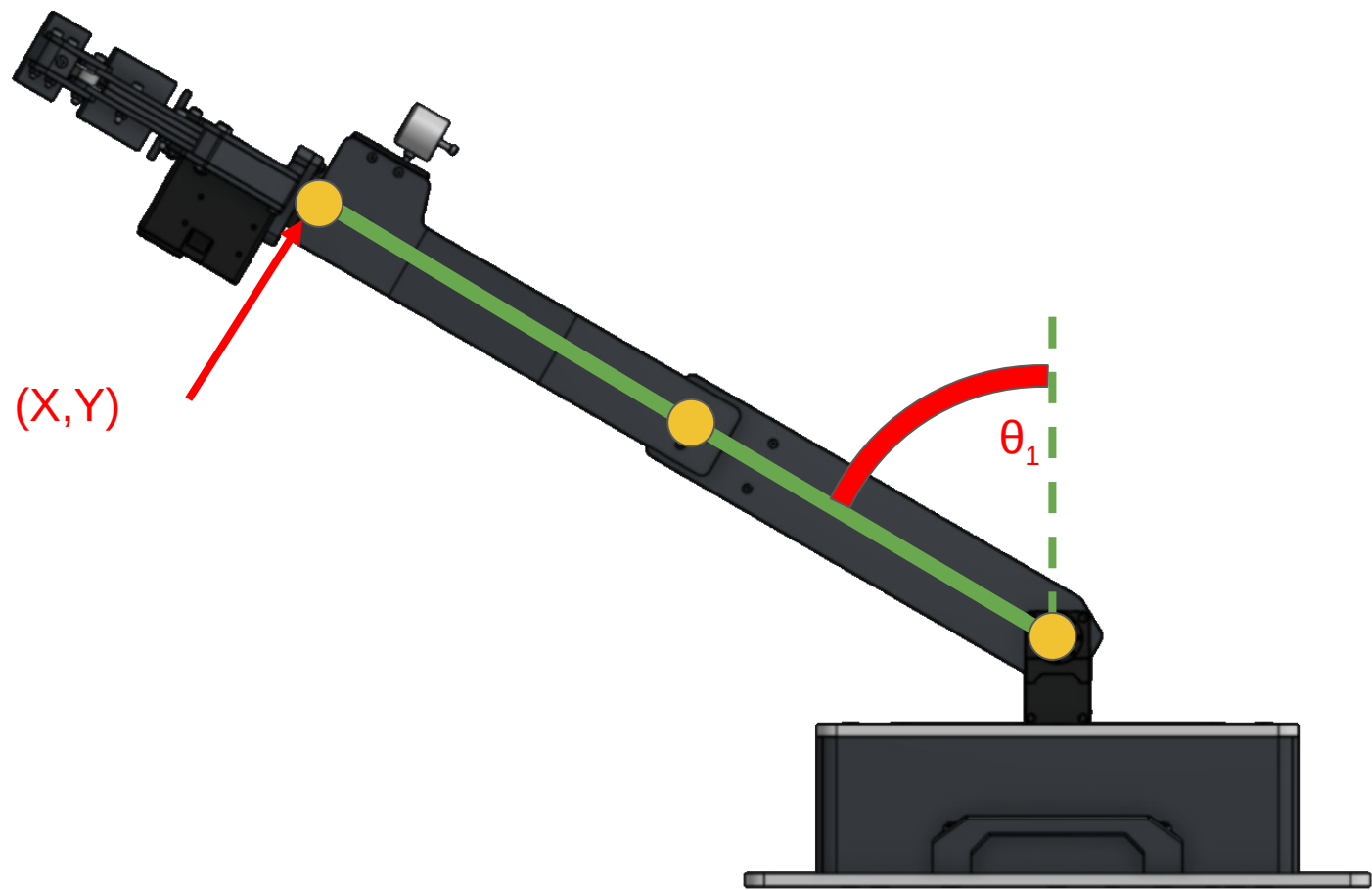
$$L_1 = 250\text{mm}$$

$$L_2 = 250\text{mm}$$

$$\theta_1 = 60^\circ$$

$$\theta_2 = 0^\circ$$

$$(X, Y) = (?, ?)$$

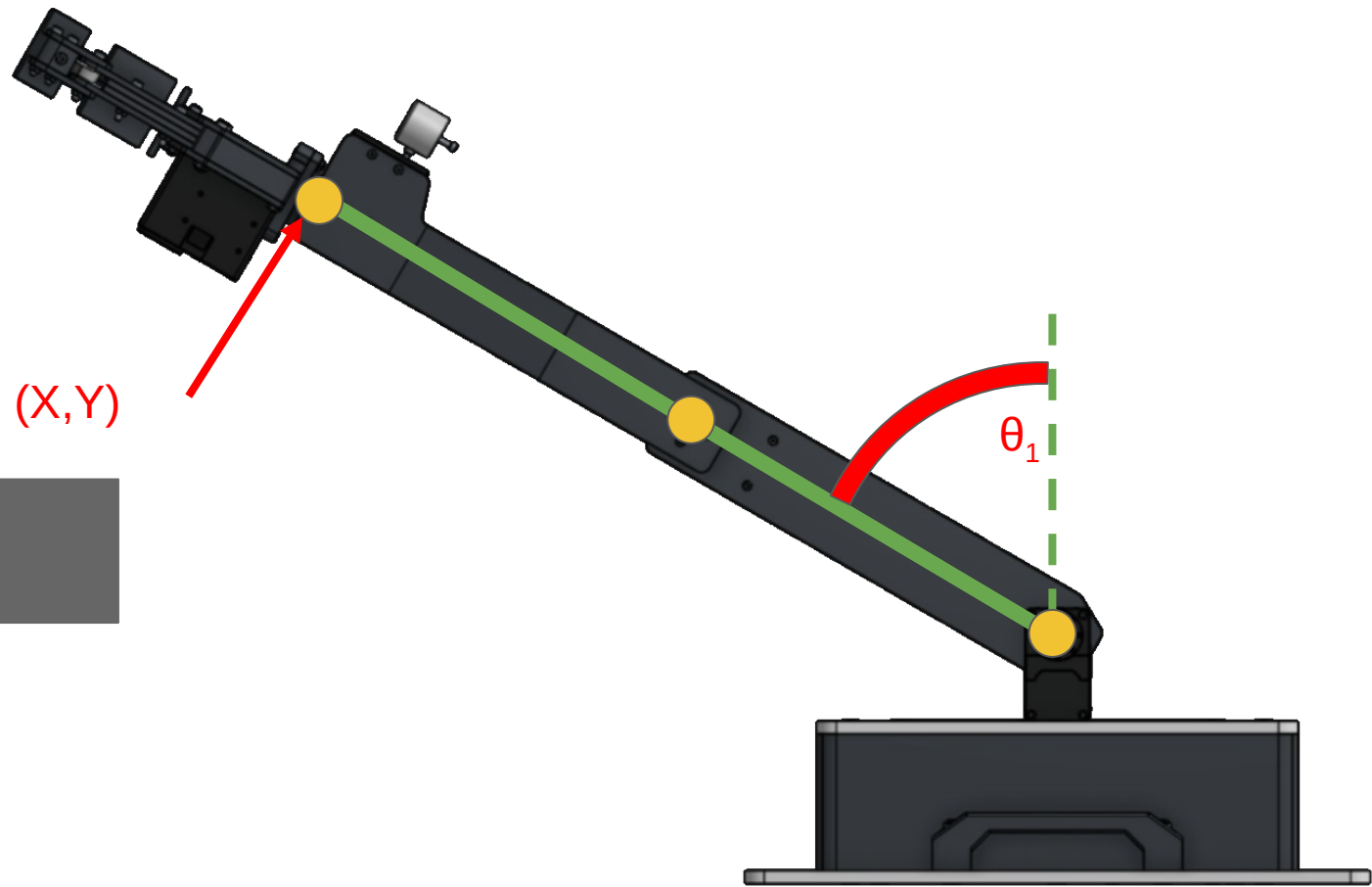


Try It Yourself 2:

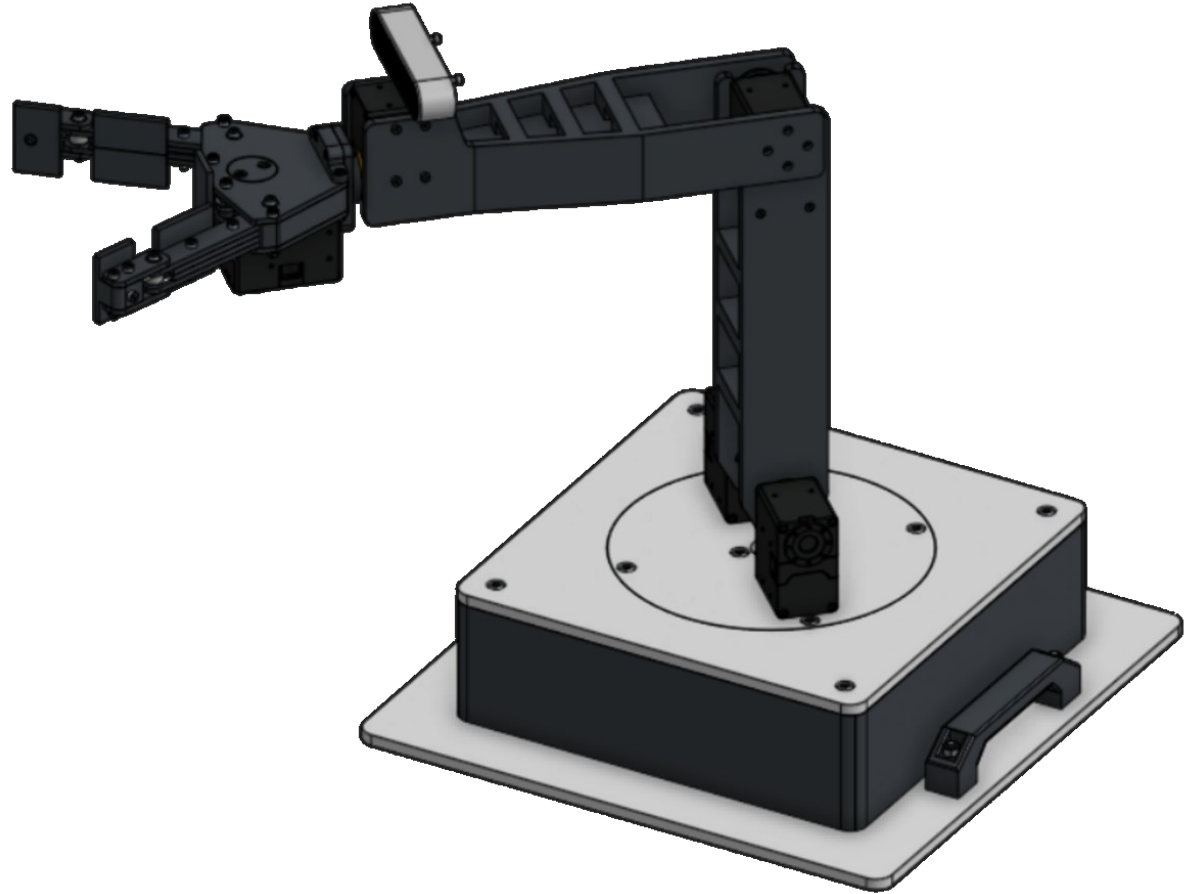
$L_1=250\text{mm}$
 $L_2=250\text{mm}$
 $\theta_1=60^\circ$
 $\theta_2=0^\circ$
 $(X,Y) = (?,?)$

Solution:

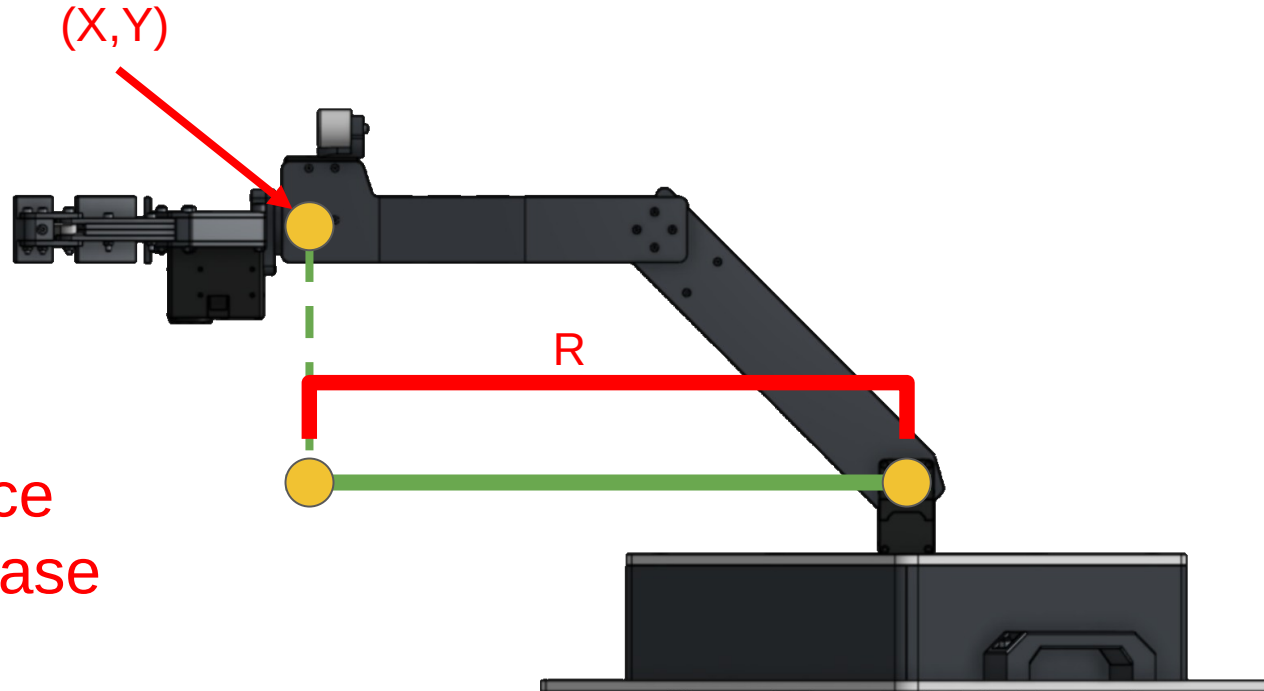
Click To Reveal



Forward Kinematics With Base Rotation



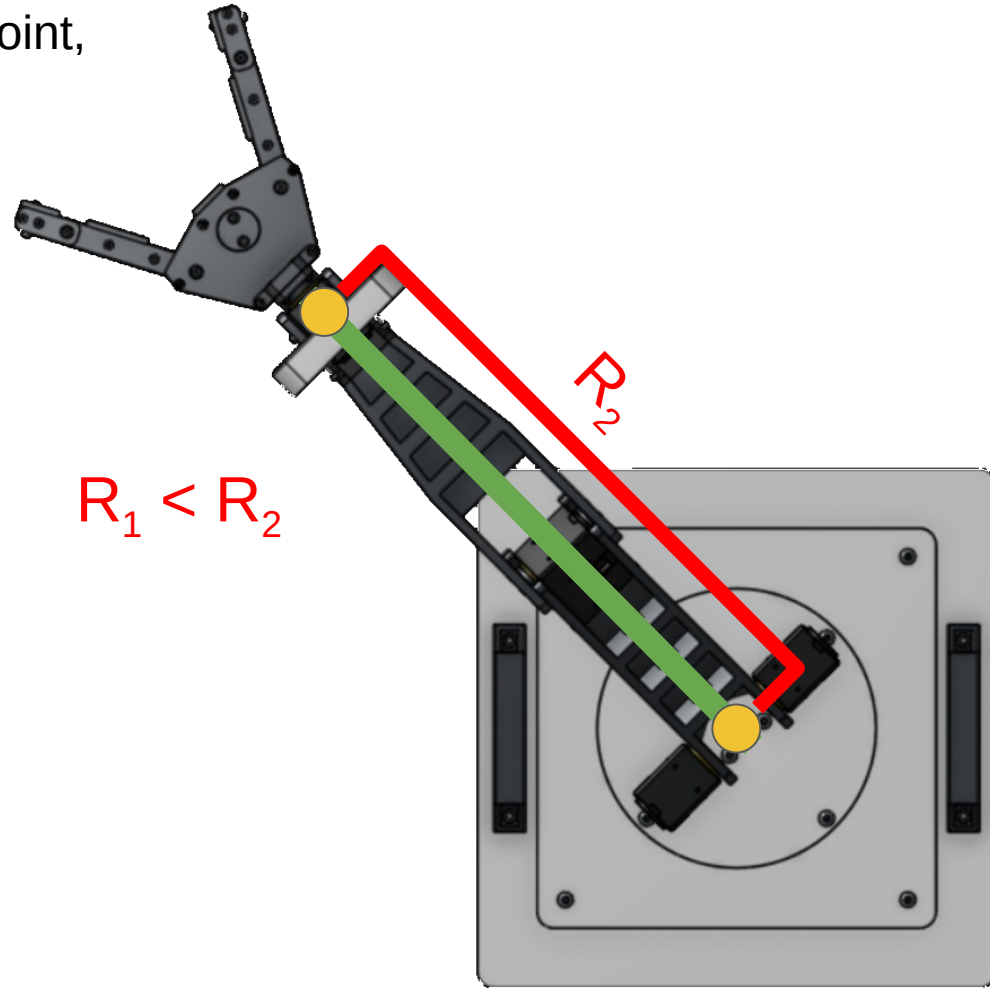
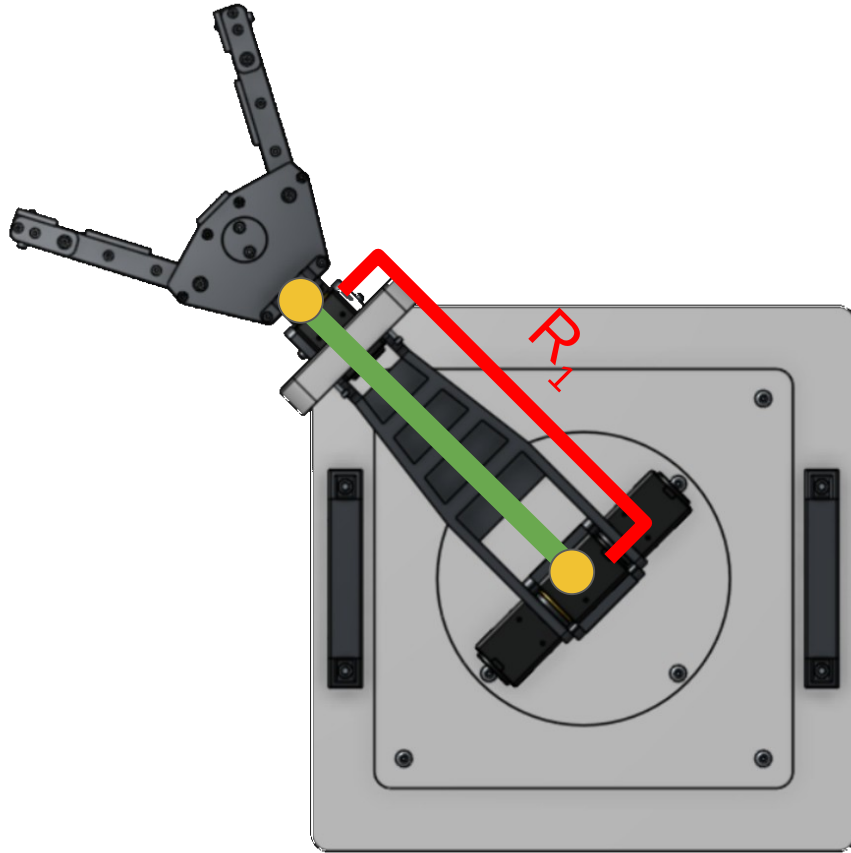
The arm's forward kinematics with base rotation is dependent on rotation around the center point of the base



Where

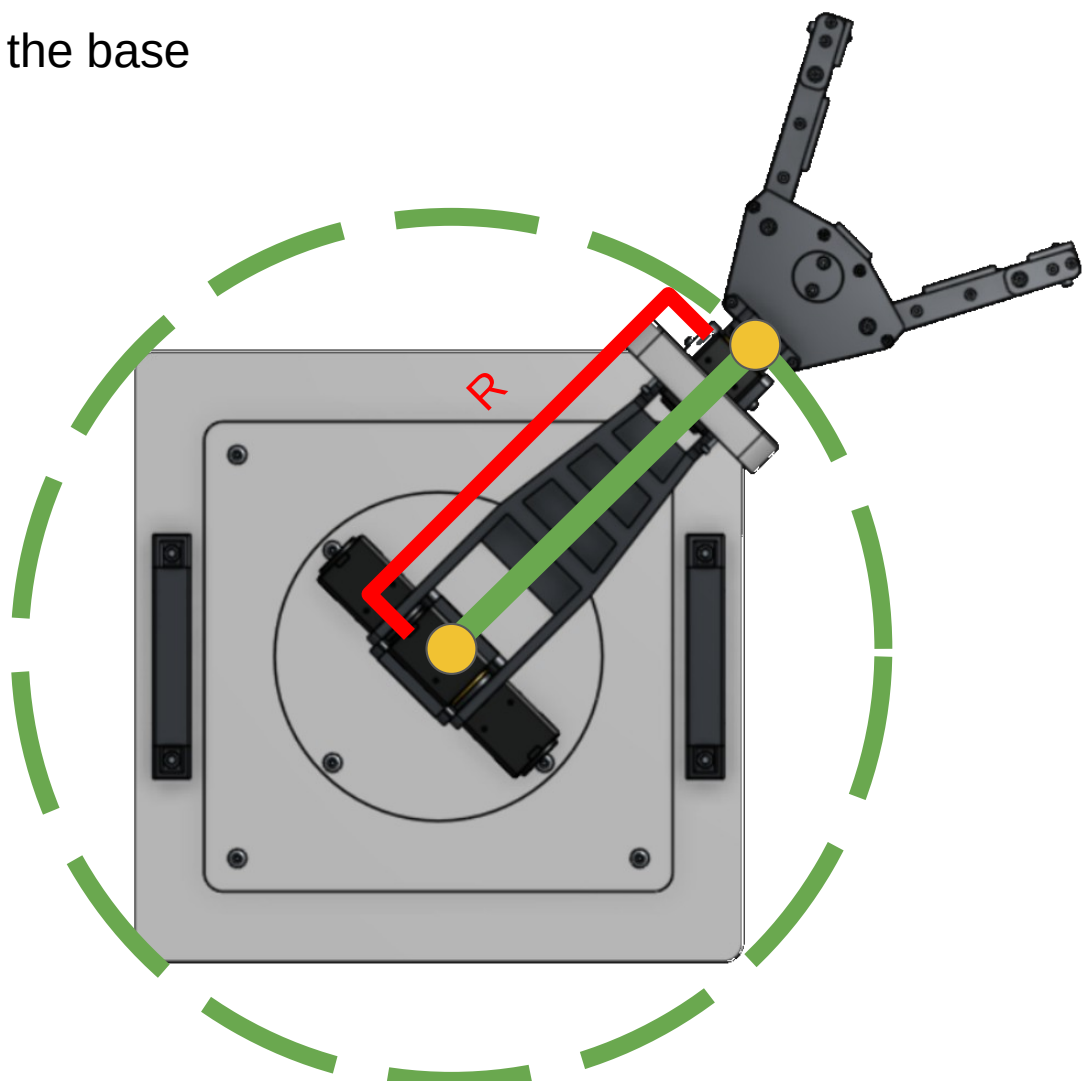
R = Horizontal distance
from end effector to base
center point

R is variable because the end effector can move away or closer to the base center point, horizontally

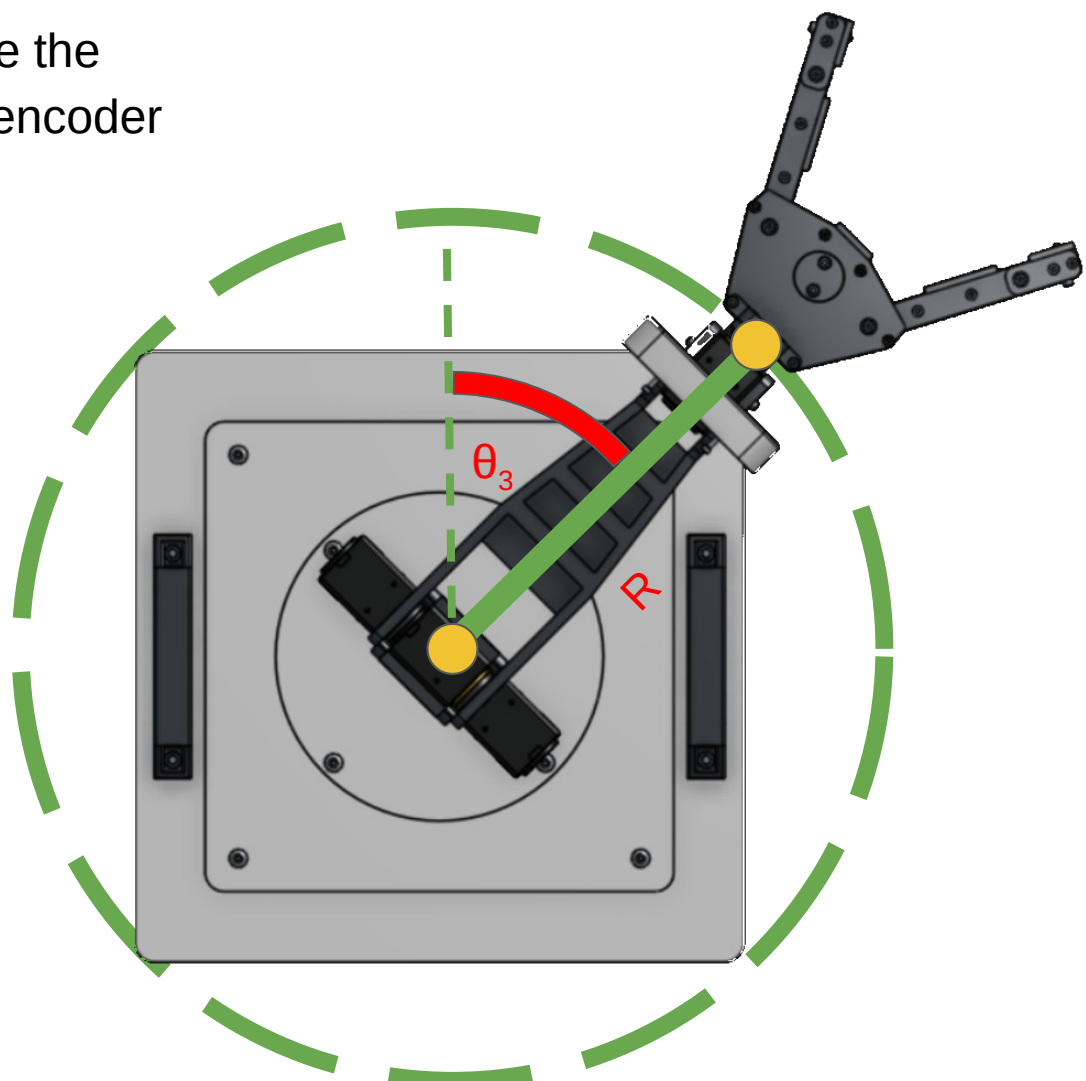


$$R_1 < R_2$$

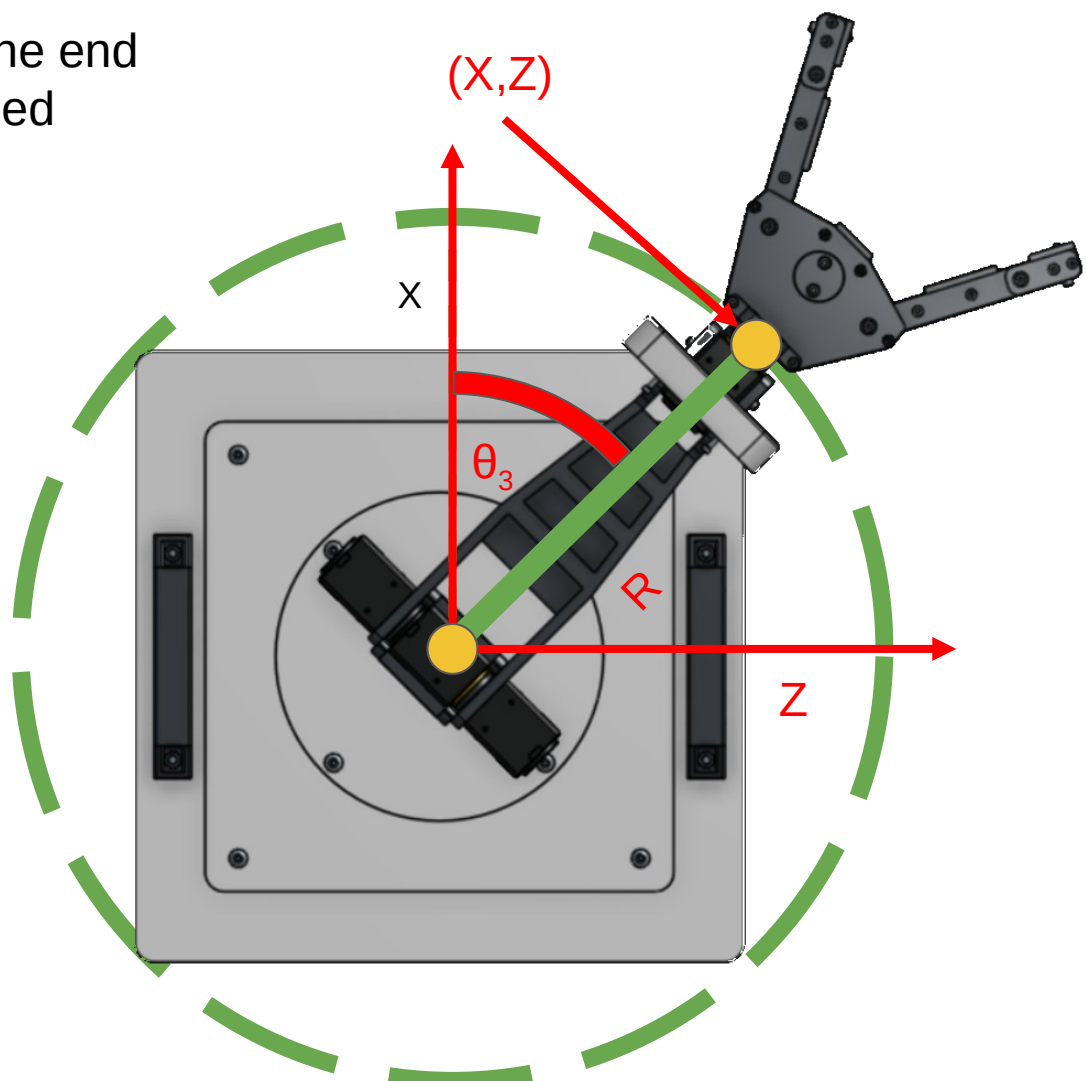
R forms a circle around the center of the base
when rotated



θ_3 is known because we can measure the rotation of the base actuator with its encoder



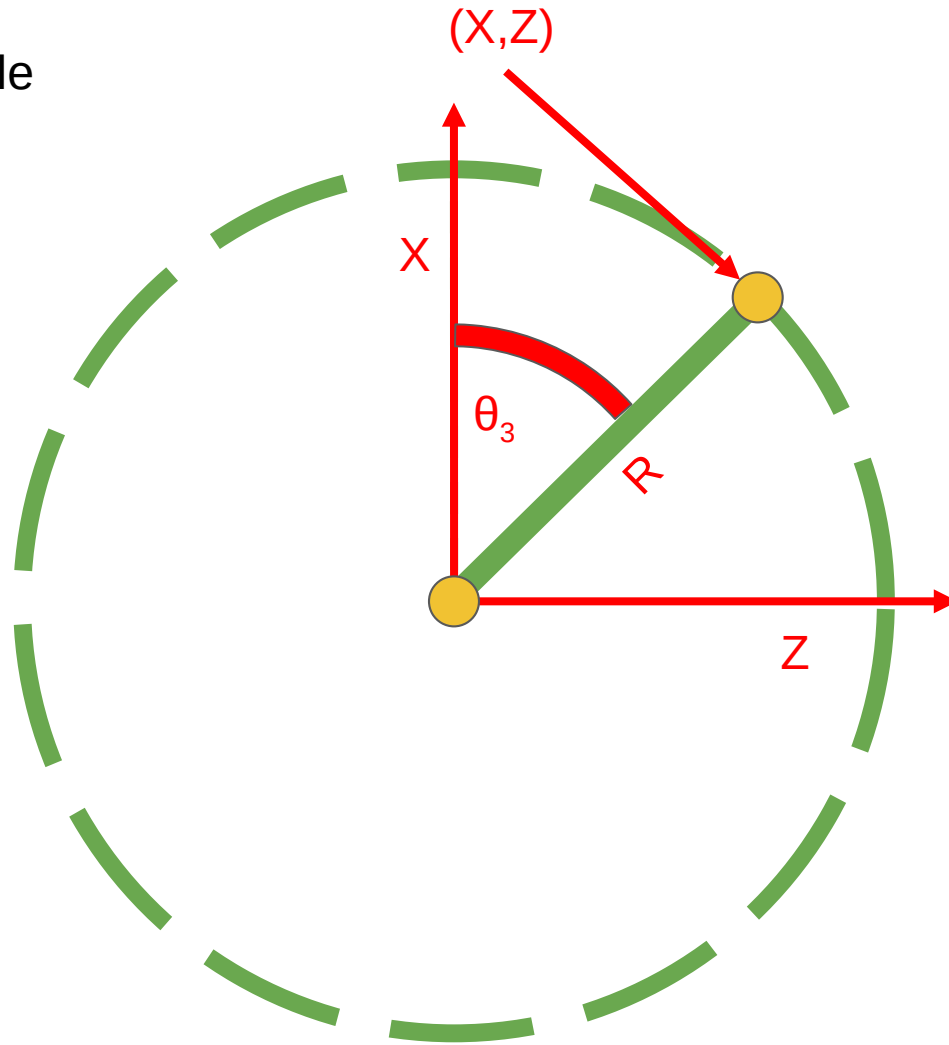
So if R is calculated, the position of the end effector in 3D space can be determined



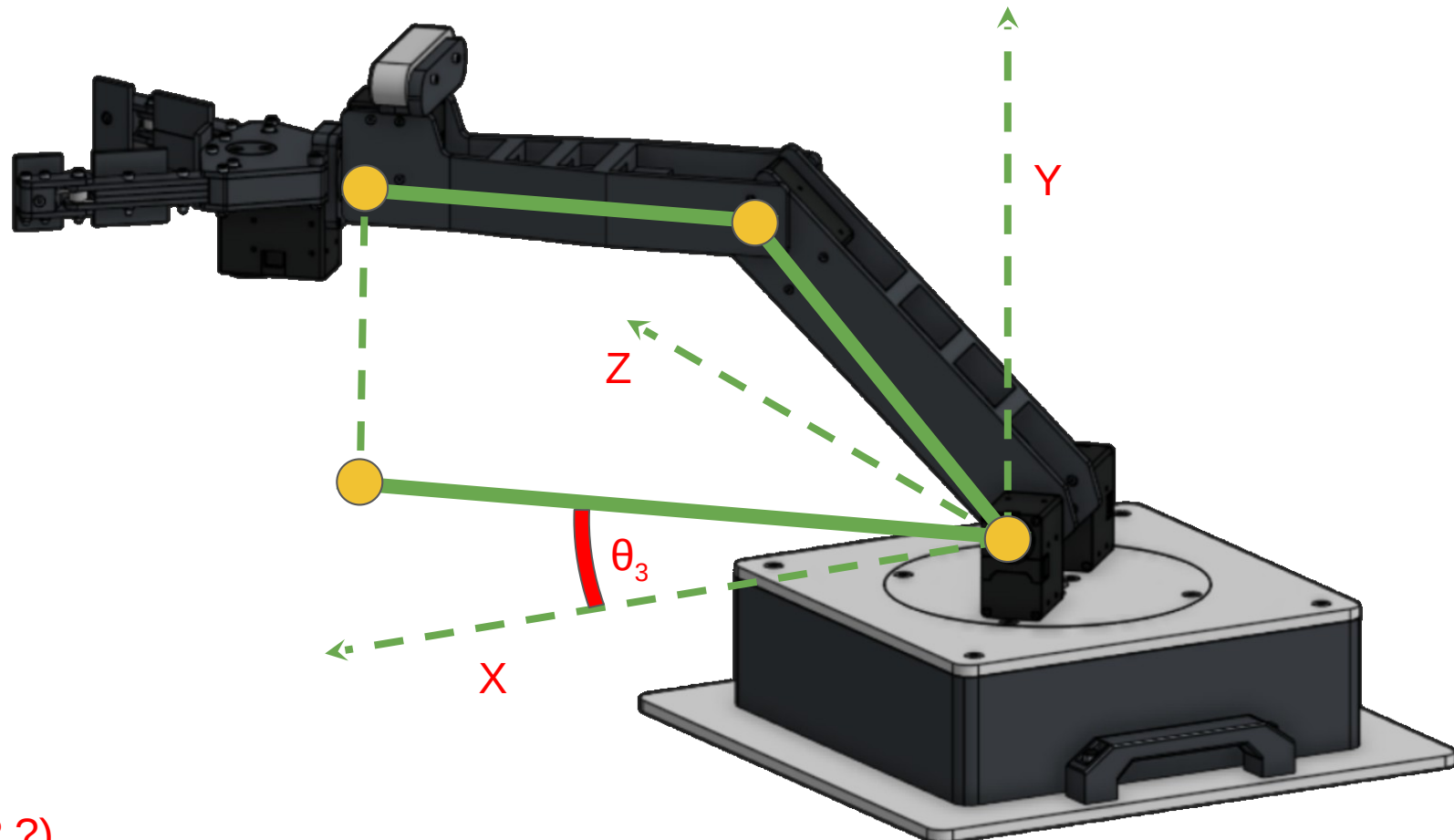
The problem can be represented as a circle with a radius

To find (X,Z)

1. Find R
2. Calculate the location of R on the circle



Example Problem

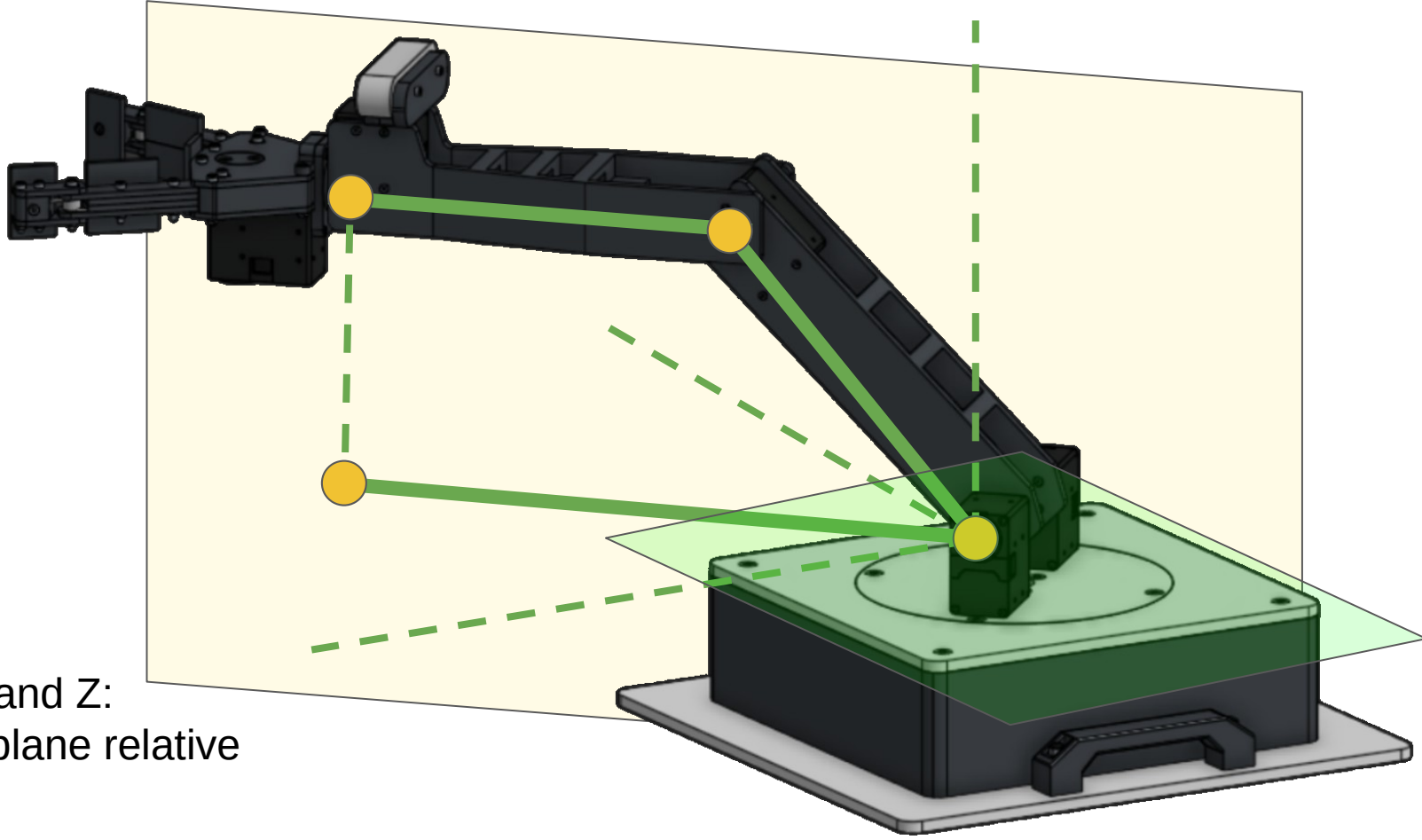


$L_1=250\text{mm}$
 $L_2=250\text{mm}$
 $\theta_1=45^\circ$
 $\theta_2=45^\circ$
 $\theta_3=45^\circ$
 $(X,Y,Z) = (?,?,?)$

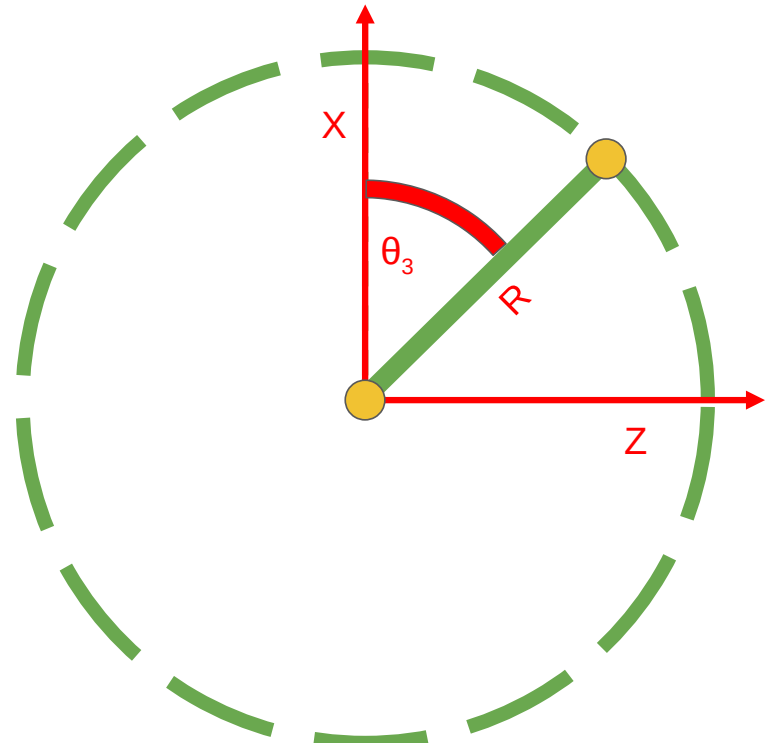
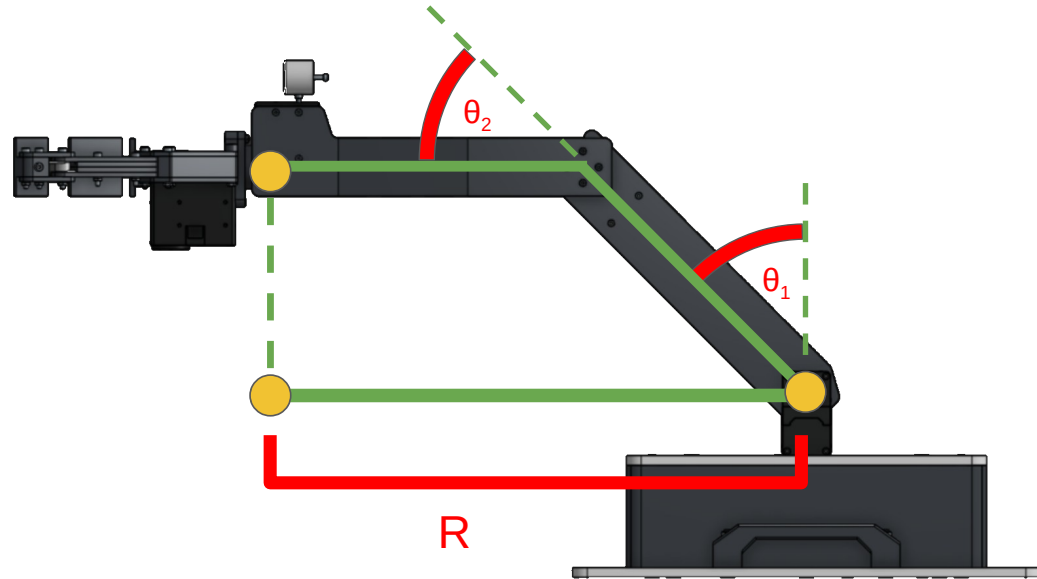
This problem can be split
into two 2D problems

Solving for R:
along the
XZ Plane

Solving for X and Z:
Along an XY plane relative
to the arm

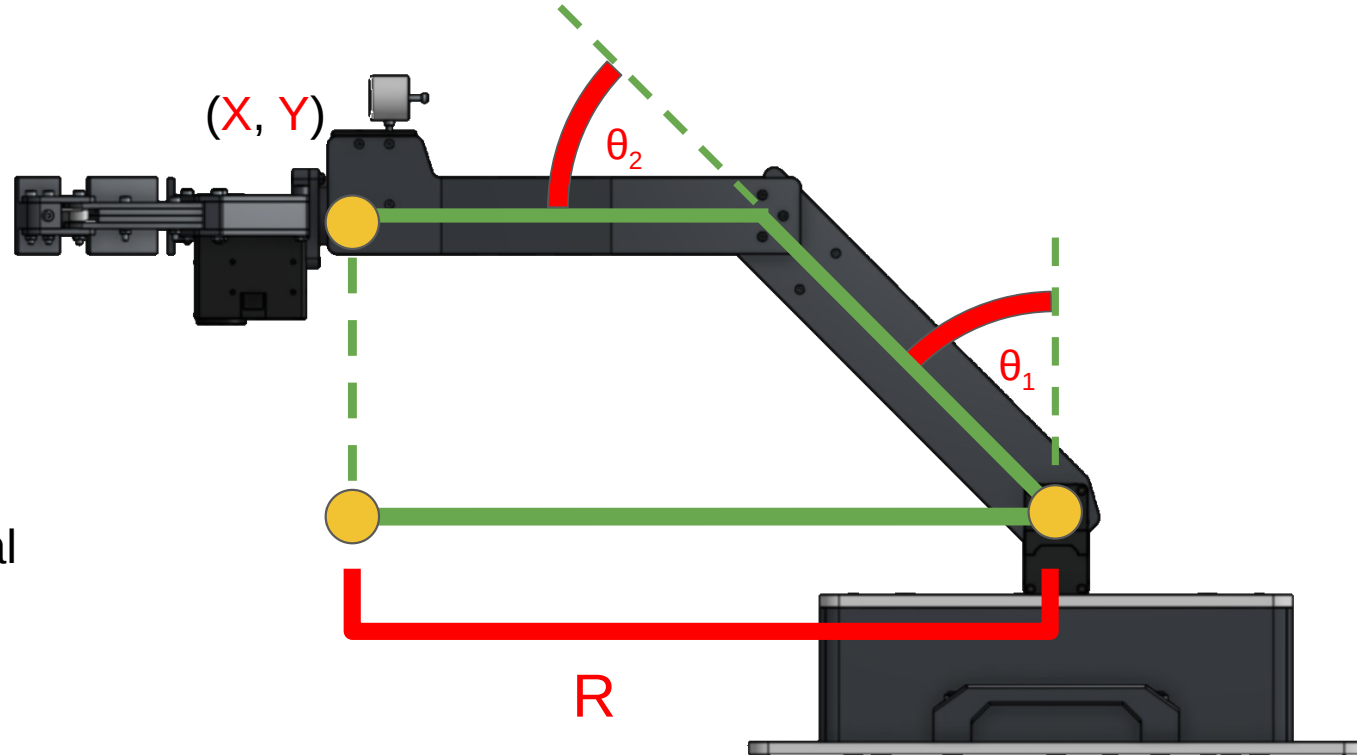


The problem can be solved
along these two planes



R will be equal to the X
value we calculated when
analyzing this position in
2D

For this problem
($L_1=250\text{mm}$, $L_2=250\text{mm}$,
 $\theta_1=45^\circ$, $\theta_2=45^\circ$)
We solved it already and
determined (X,Y) to equal
(426.777 , 176.777)
In this frame, $R=x$
Therefore, $R=426.77\text{mm}$
and $Y=176.78\text{mm}$

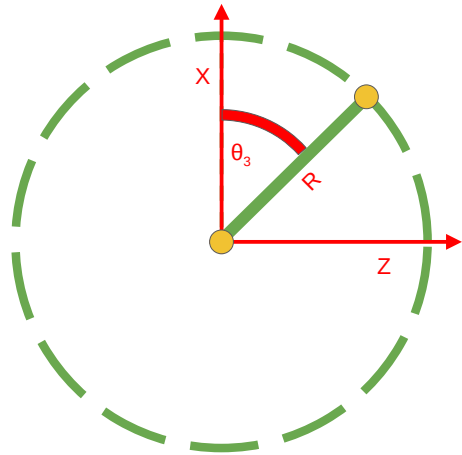
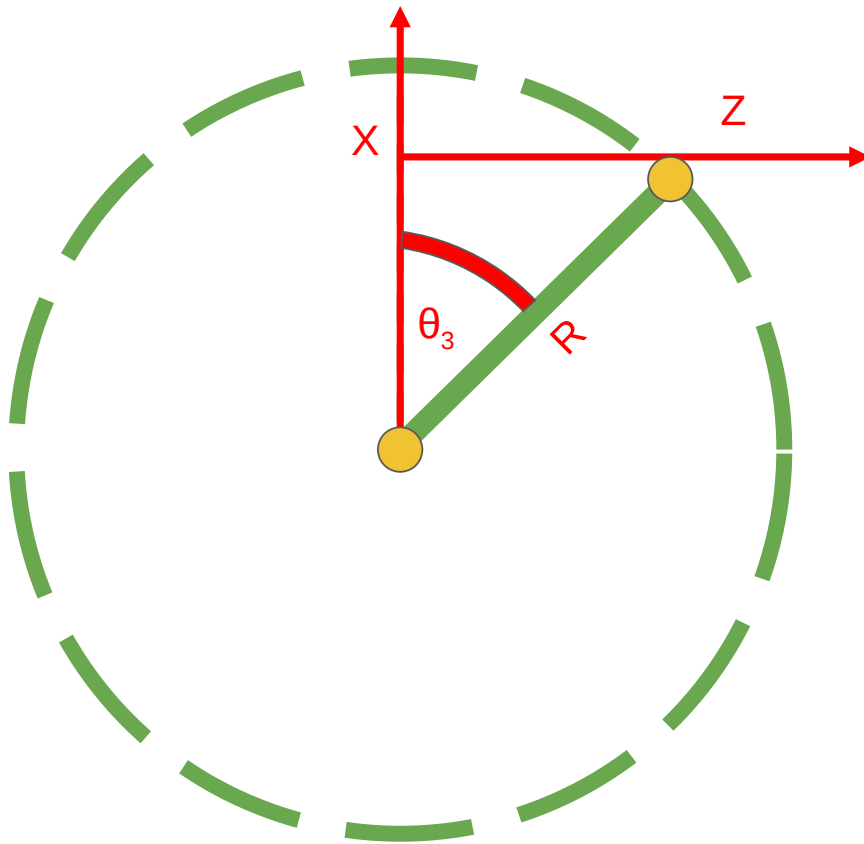


Along the XZ Plane

$R = 426.77\text{mm}$
 $\theta_3 = 45^\circ$

$Z = R\sin(\theta_3)$
 $X = R\cos(\theta_3)$

$Z = 301.77\text{mm}$
 $X = 301.77\text{mm}$



Example Problem: Solution

$L_1=250\text{mm}$

$L_2=250\text{mm}$

$\theta_1=45^\circ$

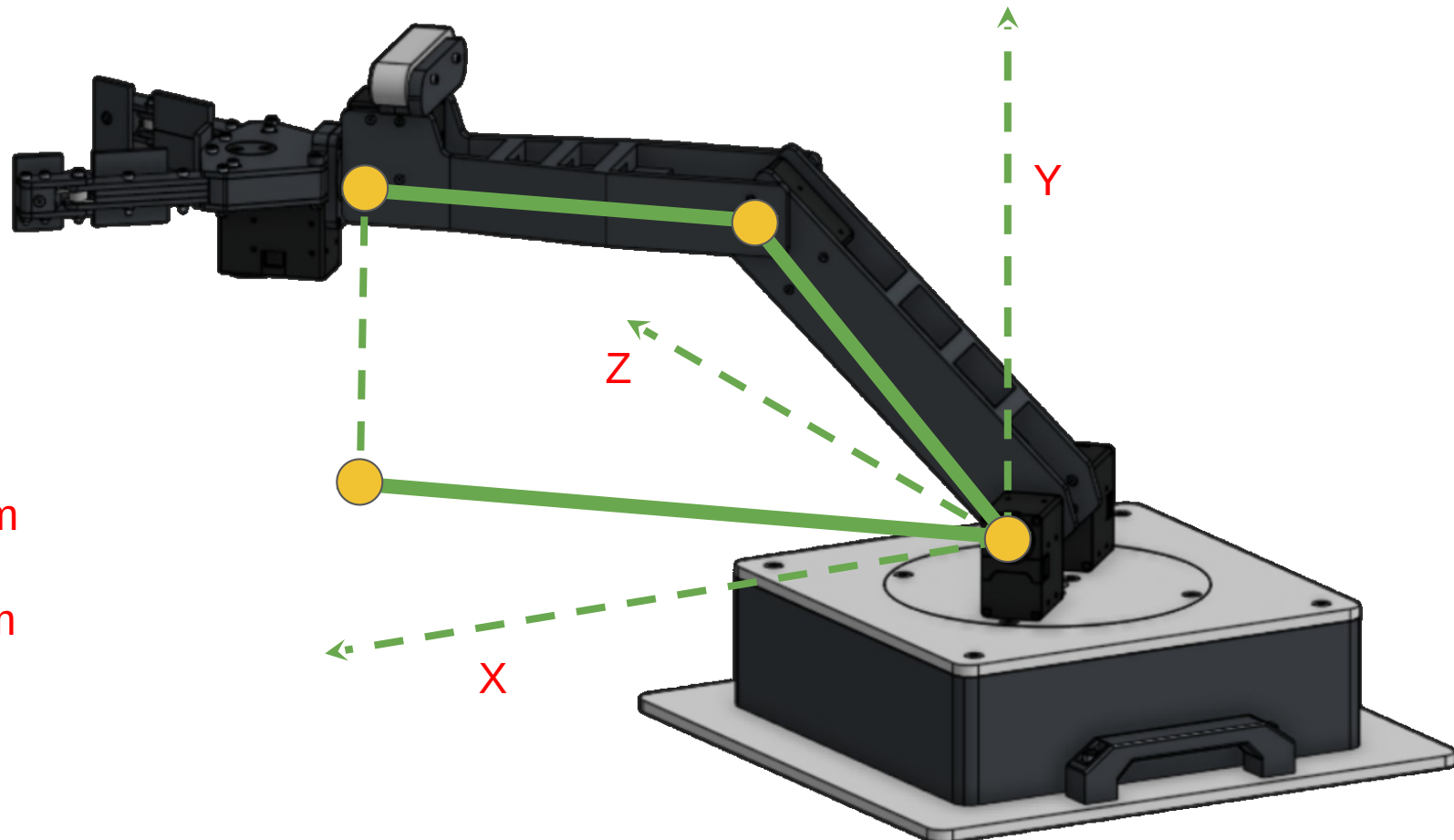
$\theta_2=45^\circ$

$\theta_3=45^\circ$

$X = 301.77\text{mm}$

$Y=176.78\text{mm}$

$Z = 301.77\text{mm}$



Practice Problem

$L_1=250\text{mm}$

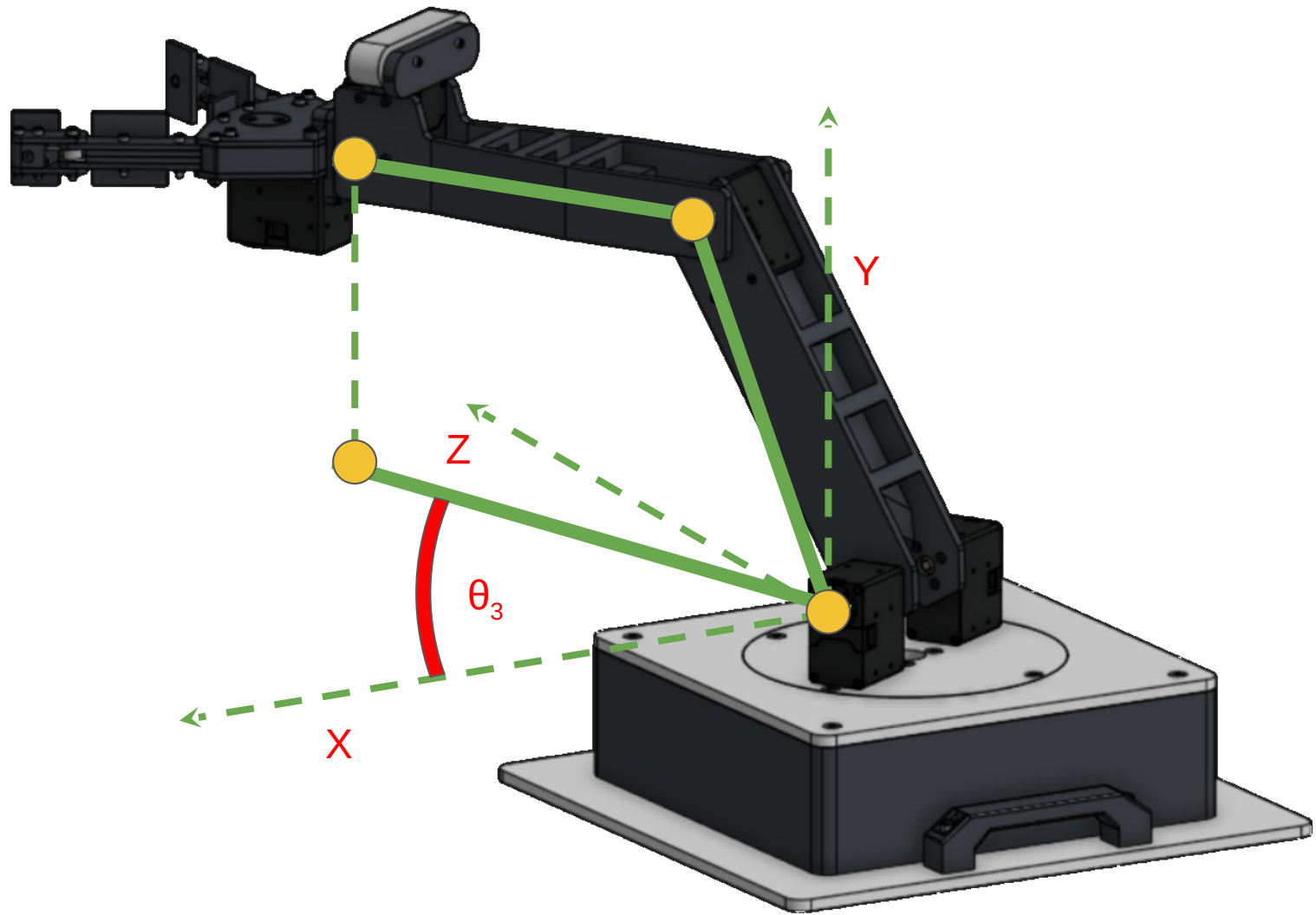
$L_2=250\text{mm}$

$\theta_1=30^\circ$

$\theta_2=30^\circ$

$\theta_3=60^\circ$

$(X,Y,Z) = (?, ?, ?)$



Practice Problem

$L_1=250\text{mm}$

$L_2=250\text{mm}$

$\theta_1=30^\circ$

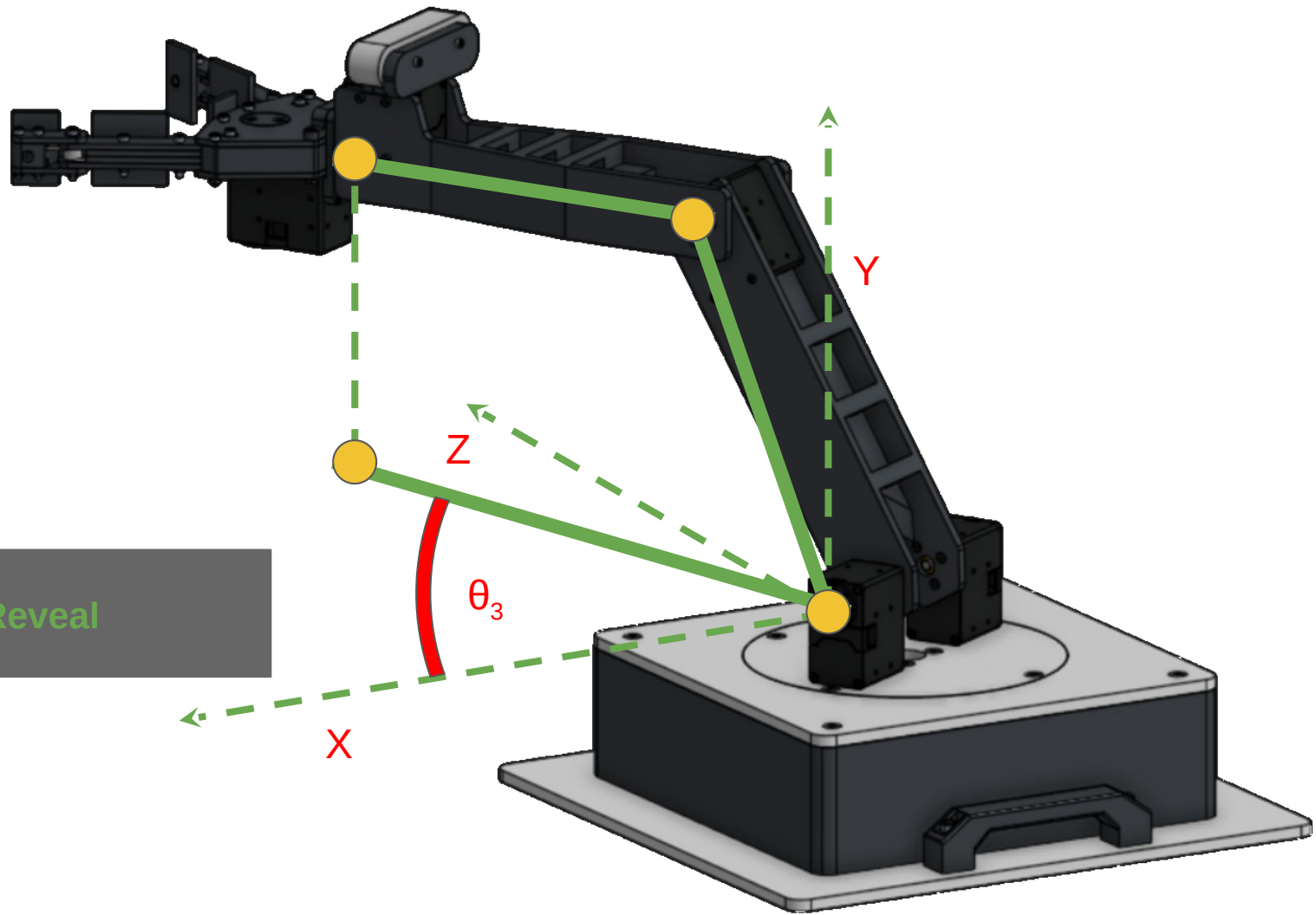
$\theta_2=30^\circ$

$\theta_3=60^\circ$

$(X,Y,Z) = (?, ?, ?)$

Solution:

Click To Reveal



Practice Problem 2

$L_1=250\text{mm}$

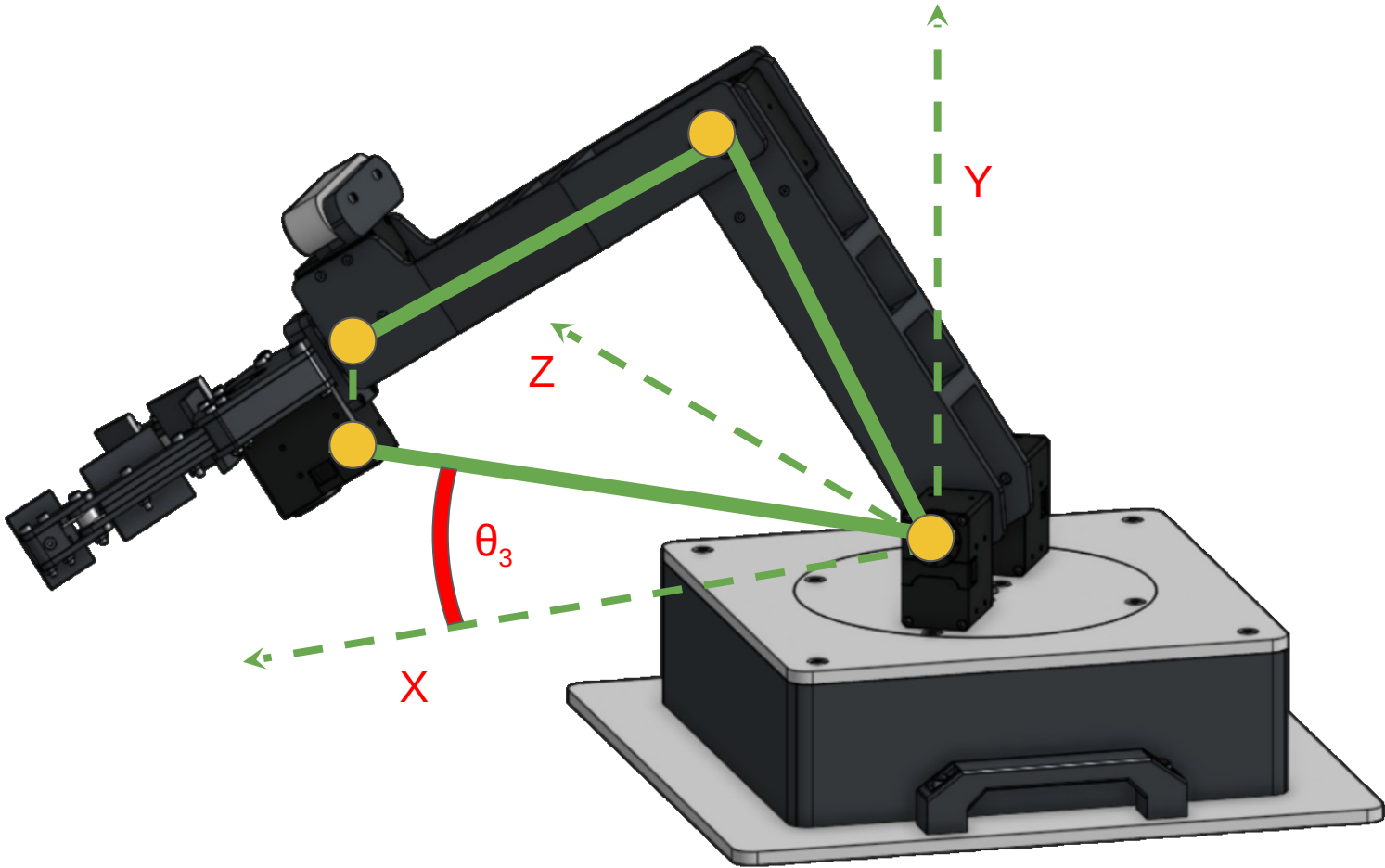
$L_2=250\text{mm}$

$\theta_1=30^\circ$

$\theta_2=90^\circ$

$\theta_3=30^\circ$

$(X,Y,Z) = (?, ?, ?)$



Practice Problem 2

$L_1=250\text{mm}$

$L_2=250\text{mm}$

$\theta_1=30^\circ$

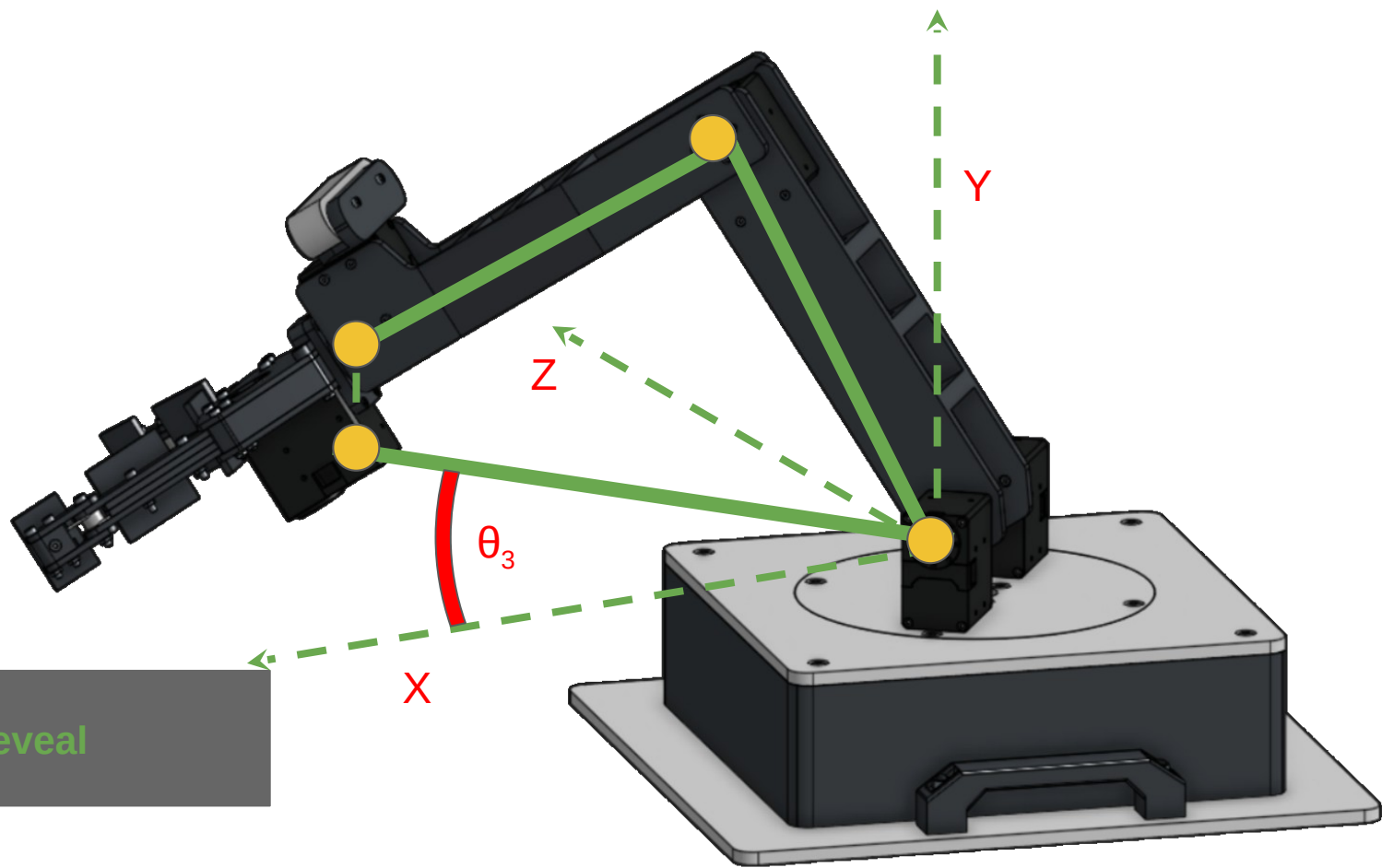
$\theta_2=90^\circ$

$\theta_3=30^\circ$

$(X,Y,Z) = (?, ?, ?)$

Solution:

Click To Reveal



Desmos Calculator

<https://www.desmos.com/3d/9bnvqc263x>

